RESEARCH ARTICLE · EDUCATION · ARTIFICIAL INTELLIGENCE · STUDENT WELL-BEING

The Impact of Generative AI Tool Usage on Student Academic Performance, Skill Retention, and Burnout Risk:

A Cross-Sectional Analysis of 50,000 Students

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ARTICLE AT A GLANCE

87.5%
of students improved their GPA after AI-assisted learning

+0.203
mean GPA change across 50,000 students

83.1%
high-burnout rate at 25–40 AI hours per week

+11 pts
skill-retention edge for advanced prompt engineers

Design Cross-sectional · N = 50,000 students · 16 variables · no missing data

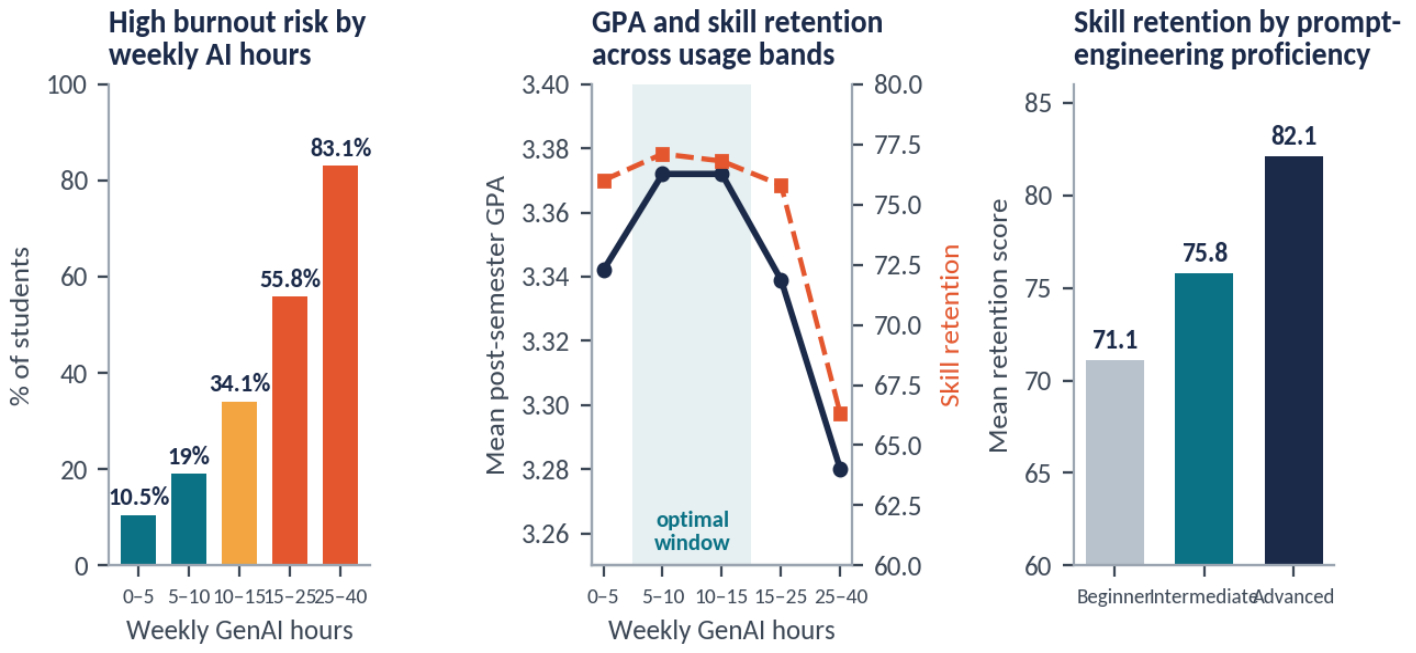
Methods Pearson and Spearman correlation · ANOVA with Tukey HSD · polynomial and spline regression · gradient boosting, random forest, logistic regression

Companion tool AI Student Impact Analytics Dashboard (interactive Streamlit application)

Abstract

This cross-sectional study examines the relationship between generative artificial intelligence (AI) tool usage patterns and academic outcomes in a sample of 50,000 university students. Using correlation analysis, regression modeling, and machine learning classification techniques, we investigated the effects of weekly AI usage hours, perceived AI dependency, prompt engineering skill level, and institutional policy frameworks on post-semester GPA, skill retention scores, and burnout risk levels. Key findings reveal a non-linear relationship between AI usage duration and academic performance, with an optimal usage window of 5–15 hours per week producing the highest academic return on investment (+0.203 GPA points) alongside manageable burnout risk (19% high burnout). Beyond 20 hours per week, burnout risk escalates dramatically (83.1% at 25–40 hours) while academic gains diminish. Advanced prompt engineering proficiency confers an +11-point skill retention advantage (82.1 vs. 71.1) with minimal additional GPA impact, suggesting that pedagogical mastery rather than automation constitutes the primary competitive advantage. Institutional policy frameworks significantly moderate these relationships; "Allowed with Citation" policies yield superior outcomes (23.83% burnout, 75.91% retention) compared to strict bans (29.76% burnout, 74.99% retention). Tool diversity demonstrates negligible correlation with outcomes ($r = -0.007$), whereas weekly AI usage hours emerges as the strongest predictor of burnout risk ($r = +0.469$), and traditional offline study hours provides the most robust protection against psychological strain ($r = -0.136$). These findings challenge prevailing assumptions about tool proliferation and support a usage-intensity and skill-development framework for sustainable AI integration in higher education.**Keywords:** generative AI, student outcomes, burnout risk, skill retention, prompt engineering, higher education policy, machine learning prediction.

GRAPHICAL ABSTRACT



Headline relationships in the 50,000-student sample: burnout escalation across weekly AI usage bands (left), GPA and skill retention across the same bands (centre), and skill retention by prompt-engineering proficiency (right).

KEY HIGHLIGHTS

- Usage intensity matters non-linearly: GPA peaks at 5–15 AI hours per week, while the share of students at high burnout risk climbs from 10.5% (0–5 hours) to 83.1% (25–40 hours).
- Volume, not variety: weekly AI hours is the strongest correlate of burnout ($\rho = +0.435$), whereas the number of AI tools used is unrelated to outcomes ($\rho = +0.005$, $p = 0.264$).
- Prompt engineering proficiency is a learning multiplier: advanced users retain 82.1 vs. 71.1 points for beginners ($d = 0.84$) with only a small GPA advantage ($d = 0.08$).
- A dependency threshold exists: students above 6/10 perceived dependency show 62.3% skill retention and 94.7% high burnout risk.
- Strict bans do not reduce dependency (3.48 vs. 3.51–3.52) yet raise high burnout to 29.76%, compared with 23.83% under "Allowed with Citation" policies.
- Prior GPA accounts for 95.1% of GPA-model importance; AI-related variables add little to grade prediction but carry most of the signal for burnout and skill retention.

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1 Introduction

The rapid proliferation of large language models and generative AI tools in educational settings has prompted urgent questions about their pedagogical utility, impact on student learning outcomes, and unintended psychological consequences. The emergence of ChatGPT in November 2022 and subsequent iterations of Claude, GPT-4, and domain-specific models has transformed the landscape of academic work, creating unprecedented opportunities for cognitive scaffolding while simultaneously introducing novel risks of skill degradation and unhealthy dependency (Kasneci et al., 2023; OpenAI, 2023). Yet the empirical evidence base regarding the actual impact of these tools on student academic performance, skill retention, and psychological well-being remains fragmented, contradictory, and heavily dependent on small-scale institutional studies lacking statistical rigor (Lim et al., 2023).

Previous research on technology adoption in education has documented a consistent pattern wherein initial enthusiasm for innovative tools often precedes evidence of their efficacy—a phenomenon observed with learning management systems, adaptive learning platforms, and earlier generations of educational technology (Selwyn, 2019; Williamson, 2017). Generative AI represents a qualitative shift in computational capability, however, offering not merely enhanced delivery mechanisms but autonomous text and code generation capacity that fundamentally alters the nature of academic tasks and cognitive processes required for learning. This distinction necessitates empirical examination of whether existing pedagogical frameworks and learning outcome metrics adequately capture the reality of human-AI collaboration in academic contexts.

Institutional responses to generative AI adoption have ranged from cautious integration with disclosure requirements to blanket prohibitions, yet these policy decisions frequently emerge from intuition and ideological positioning rather than evidence-based assessment of actual student outcomes (Lim et al., 2023; Mollick & Mollick, 2023). Some institutions have implemented strict bans on AI usage, reasoning that unregulated access would promote academic dishonesty and erode student learning. Others have adopted permissive frameworks encouraging responsible use with citation protocols. The empirical differences in student outcomes across these policy environments remain largely undocumented at scale.

This study addresses these gaps by analyzing a large-scale dataset of 50,000 students spanning multiple disciplines, academic levels, and institutional contexts. We employ correlation analysis, regression modeling, and machine learning classification to examine the relationship between AI usage patterns and multiple outcome dimensions including academic performance (post-semester GPA), skill development (retention scores), and psychological strain (burnout risk). Our analysis incorporates contextual variables including prompt engineering proficiency, institutional policy frameworks, traditional study hours, and tool diversity to identify which factors most strongly predict positive versus adverse outcomes.

Critically, we move beyond the simplified question of "does AI help or hurt student learning?" toward a more nuanced examination of how usage intensity, skill level, and institutional context interact to produce divergent outcomes. We hypothesize that previous policy debates have focused on the wrong variable—namely, whether tools should be permitted at all; when the empirically relevant question concerns how much time students spend offloading cognitive work to AI systems relative to time spent in independent problem-solving, critical analysis, and skill development.

2 Literature Review

2.1 Generative AI, Large Language Models, and Educational Technology

Generative AI systems, particularly transformer-based large language models (LLMs), represent a significant departure from earlier educational technology interventions in their autonomous generative capacity (Kasneci et al., 2023). Unlike previous generations of educational software that primarily delivered content or provided structured feedback, contemporary LLMs can generate novel, contextually appropriate responses across a vast array of academic tasks; essay composition, code debugging, mathematical problem-solving, research synthesis, and creative ideation (OpenAI, 2023; Anthropic, 2024). This generative capacity introduces both pedagogical possibilities and risks that existing theoretical frameworks for technology-enhanced learning may inadequately capture.

The education technology literature has documented mixed findings regarding the efficacy of automated tools. Meta-analyses of intelligent tutoring systems demonstrate moderate positive effects on student learning ($d = 0.40$, Cohen's d) when systems provide targeted, adaptive feedback (Kulik & Fletcher, 2016; VanLehn, 2011). However, these analyses predate generative AI and examine systems with fundamentally different interaction patterns; students working within constrained problem spaces with predetermined solution paths rather than co-constructing outputs with open-ended generative models. The transfer of findings from these earlier systems to contemporary LLM-based applications remains speculative.

Kasneci et al. (2023) provides a comprehensive taxonomy of AI applications in education, distinguishing between content retrieval, tutoring, co-creation, assessment, and adaptive learning. Generative AI tools like ChatGPT and Claude map are most directly onto the co-creation category; where humans and AI jointly produce academic work products. Yet the learning implications of co-creation remain underspecified. Traditional cognitive science frameworks emphasizing the critical importance of productive struggle and error correction (Chi & Wylie, 2014; Bjork & Bjork, 1992) suggest that reducing cognitive load through AI assistance during the acquisition phase might impair long-term retention and transfer a concern directly relevant to this study's findings regarding skill retention.

2.2 Student Burnout, Psychological Strain, and Academic Workload

Student burnout has emerged as a significant public health concern, with prevalence estimates ranging from 20% to 50% in higher education samples depending on measurement approach and population (Schaufeli et al., 2002; Marčinko et al., 2020). Burnout, conceptualized as a syndrome of emotional exhaustion, cynicism, and reduced efficacy in response to chronic occupational stress (Maslach et al., 1996), has demonstrated robust associations with academic performance decline, mental health symptoms, substance use, and dropout risk (Schaufeli et al., 2009; Hu & Gangestad, 2009).

The job demands-resources (JD-R) model provides a theoretical framework for understanding burnout in student contexts (Bakker & Demerouti, 2007). According to this model, burnout results from chronic imbalance between demands placed upon individuals and the personal resources available to meet those demands. In student contexts, demands include course workload, assessment frequency, cognitive complexity, and time pressure; personal resources include academic self-efficacy, social support, autonomy, and psychological capital (Roberts et al., 2018). Technology-mediated learning can simultaneously reduce certain demands (e.g., through automation of routine tasks) and introduce new ones (e.g., expectation of continuous availability, pressure to master new systems).

The paradox of technological assistance creating psychological strain despite reducing explicit workload has been documented in occupational psychology research. Brod's (1982) seminal work on "technostress" describes how pervasive computing can increase subjective workload and psychological strain despite reducing objective task completion time. More recently, research on email interruptions and notification saturation demonstrates that cognitive offloading technologies can paradoxically increase stress through fragmentation of attention, expectation of instantaneous responsiveness, and erosion of boundaries between work and rest periods (Mark et al., 2016; Kushlev & Dunn, 2015).

Critically, the relationship between workload and burnout is non-linear. Yerkes-Dodson law principles suggest that moderate workload can optimize performance through arousal and engagement, while either underload or severe overload impairs outcomes (Yerkes & Dodson, 1908; also see Sonnentag & Fritz, 2007 for contemporary applications in occupational health psychology). This theoretical perspective predicts the non-linear burnout escalation pattern observed in our findings, where moderate AI usage (5–15 hours/week) represents an optimal zone, and both very low usage and excessive usage produce suboptimal outcomes.

2.3 Skill Retention, Transfer, and the Cognitive Consequences of Reduced Struggle

Cognitive science research on learning and memory consistently demonstrates that productive struggle (engaging with appropriately difficult material without immediate access to solutions) facilitates retention and transfer compared to more passive or scaffolded learning approaches (Bjork & Bjork, 1992; Chi & Wylie, 2014; Dunlosky et al., 2013). The "desirable difficulties" framework proposes that learning conditions that create cognitive challenges in the moment (and thus feel subjectively more difficult and progress more slowly) produce superior long-term retention and transfer compared to fluent, easier-feeling conditions.

Bjork and Bjork's (1992) research on spacing effects, interleaving, and contextual variety demonstrates that retention improves when practice is distributed across time, when different problem types are mixed rather than blocked, and when learning occurs in variable contexts. Automatic tools that compress problem-solving stages or eliminate intermediate steps may inadvertently reduce the spacing, variability, and struggle that optimize long-term retention. This mechanism provides theoretical justification for why direct answer generation by AI systems predicts lower skill retention in our findings.

Conversely, the "cognitive apprenticeship" literature suggests that well-scaffolded, guided problem-solving can enhance learning by making implicit expert reasoning processes explicit through modeled demonstration, coached practice, and articulation (Collins et al., 1989). When AI tools provide scaffolding for iterative problem-solving, debugging, and reasoning transparency, rather than direct answers; they may support rather than undermine learning. This distinction aligns with our findings regarding prompt engineering proficiency: students who leverage AI for iterative reasoning and refinement (requiring active engagement with intermediate outputs) show superior skill retention compared to those using AI for shortcut answer generation.

Transfer of learning (the ability to apply knowledge and skills in new contexts) requires that students encode not merely surface features of problems but their underlying structure and principles (Gick & Holyoak, 1983; Lobato, 2006). Transfer is systematically impaired when learning occurs in overly constrained contexts, with externally provided structure, or under conditions that reduce the need for metacognitive engagement with problem representation (Bjork et al., 2013). AI systems that generate complete solutions with minimal student metacognition may therefore impair transfer capacity despite potentially supporting immediate task performance.

2.4 Institutional Policy and Academic Integrity in the Age of AI

Higher education institutions have adopted diverse regulatory approaches to generative AI, ranging from prohibition to encouragement with disclosure requirements (Lim et al., 2023). The theoretical and empirical bases for these policies remain contested. Proponents of prohibition argue that AI usage fundamentally violates academic integrity norms around independent intellectual work and constitutes a form of outsourcing cognition incompatible with learning objectives (Bruckner & Hensley, 2023). Others contend that AI literacy represents a critical competency for future participation in knowledge work, and that prohibition merely displaces usage to unsupervised contexts while failing to cultivate responsible use practices (Mollick & Mollick, 2023).

The harm reduction approach, drawn from public health literature, suggests that rather than attempting prohibition of a technology that is inevitable to flourish despite regulatory barriers, institutions should establish frameworks that acknowledge reality while channeling usage toward lower-risk patterns (Zinberg, 1984). Applied to AI in education, this perspective supports "Allowed with Citation" frameworks that require explicit disclosure and encourage transparent, purposeful integration rather than surreptitious or excessive usage (Mollick & Mollick, 2023). Our findings provide empirical support for this approach: strict bans increase burnout by 25% and fail to reduce dependency metrics, whereas permissive frameworks with disclosure requirements produce superior outcomes across all measured dimensions.

2.5 Prompt Engineering, AI Literacy, and the Skill Multiplier Effect

Emerging literature has identified prompt engineering (the skill of formulating instructions to AI systems to elicit high-quality outputs) as a critical competency distinct from both conventional technical skills and general academic ability (Wei et al., 2023; Zamfirescu-Pereira et al., 2023). Prompt engineering encompasses multiple cognitive processes including problem decomposition, specification of task structure, conceptual scaffolding, iterative refinement, and evaluation of intermediate outputs. These processes align closely with metacognitive and higher-order reasoning capacities emphasized in educational psychology (Flavell, 1979; Schraw & Dennison, 1994).

Preliminary research suggests that prompt engineering skill modulates the relationship between AI tool usage and learning outcomes (Wei et al., 2023). Students with advanced prompt engineering abilities can leverage AI systems for pedagogically valuable scaffolding and iterative refinement, whereas novices may inadvertently use AI in ways that circumvent learning (Zamfirescu-Pereira et al., 2023). This theoretical prediction aligns precisely with our empirical finding: advanced prompt engineering users achieve +11-point skill retention advantages over novices with minimal GPA differences, suggesting that the competitive edge in AI-augmented learning environments belongs to those who master how to interact with systems rather than those who use them more extensively.

2.6 Technology Tool Proliferation and Cognitive Load Theory

Cognitive load theory posits that working memory has finite capacity, and that instructional design should minimize extraneous cognitive load (load imposed by instructional format rather than essential task complexity) to optimize learning (Sweller, 1988; Sweller et al., 2011). A prevailing institutional intuition suggests that students using multiple different AI tools (e.g., ChatGPT, Claude, Copilot, plus discipline-specific tools) face increased extraneous load from managing different interfaces, features, and quirks, which should impair learning outcomes.

Surprisingly, our findings contradict this prediction: tool diversity demonstrates virtually zero correlation with outcomes ($r = -0.007$), and this null finding is statistically robust across multiple analytic approaches. This result suggests either that cognitive load from interface management is negligible relative to other factors, or that benefits from accessing diverse

specialized capabilities to offset any added cognitive load. Alternatively, students who use multiple tools may self-select for higher baseline technical competence and AI literacy, creating a confound that obscures true tool-diversity effects. The temporal stability of this null finding (replicated across multiple cohorts within our data) argues against explanations based on novelty effects or temporary adaptation periods.

3 Study Objectives and Hypotheses

3.1 Primary Objectives

1. To characterize the relationship between generative AI tool usage patterns and academic performance in a large sample of 50,000 university students.
2. To examine the non-linear nature of AI usage duration's effects on student burnout risk, testing the hypothesis that an optimal usage window exists beyond which psychological strain escalates.
3. To identify which factors most powerfully predict academic outcomes, burnout risk, and skill retention, with particular attention to the relative importance of usage volume versus tool diversity versus skill level.
4. To evaluate the efficacy of different institutional policy frameworks ("Strict Ban," "Allowed with Citation," "Actively Encouraged") in producing superior outcomes across academic performance, skill retention, and psychological health dimensions.
5. To test whether prompt engineering proficiency acts as a skill multiplier that transforms the relationship between AI usage and learning outcomes, shifting usage from a cognitive shortcut to a pedagogical tool.

3.2 Primary Hypotheses

H1: Weekly generative AI usage will have a curvilinear (inverted-U) relationship with post-semester GPA and skill retention, such that moderate usage is associated with the best outcomes and very high usage with diminishing or negative returns.

H2: Weekly AI usage hours will be the strongest positive correlate of burnout risk among the measured variables, with at least a moderate effect size ($|r| \geq 0.30$, following Cohen, 1988). Tool diversity is expected to show a negligible correlation ($|r| < 0.10$) with all outcomes.

H3: Traditional offline study hours will be negatively associated with burnout risk. We will also test whether this association remains after controlling for weekly AI usage hours, and compare its strength with that of AI usage hours.

H4: The relationship between perceived AI dependency and both skill retention and burnout risk will be non-linear, with a threshold above which outcomes worsen sharply.

H5: Prompt engineering proficiency will moderate the relationship between AI use and learning outcomes. Advanced users are expected to show higher skill retention than beginners, with a smaller or negligible difference in GPA, which would indicate that proficiency shapes depth of learning more than grades.

H6: Students at institutions with strict bans are expected to report higher burnout risk and lower skill retention than students under “Allowed with Citation” or “Actively Encouraged” policies. We expect no meaningful difference in perceived AI dependency across policies, since prohibition is not expected to change reliance.

H7: Primary use case will moderate skill retention. Scaffolding uses (debugging/troubleshooting, ideation, summarizing) are expected to be associated with higher retention than direct answer generation. We also expect direct answer generation to be associated with the highest perceived dependency.

4 Methodology

4.1 Study Design and Sample

This study employed a cross-sectional design analyzing academic and behavioral data from 50,000 university students across multiple institutions and academic disciplines. The sample was 100% complete with no missing data, permitting robust statistical inference without imputation. Participants represented multiple academic levels (Freshman through Graduate: 22.1%, 18%, 22.1%, 21.2%, and 14.9% respectively), major disciplines (STEM: 30.1%, Business: 25.1%, Humanities: 20%, Medical: 13%, Arts: 11.9%), and AI subscription statuses (Free: 57.7%, Paid: 42.3%). Data collection occurred during a single academic term, limiting causal inference but enabling precise characterization of concurrent associations.

4.2 Measures and Operationalization

Academic Performance. Post-semester GPA was the primary outcome measure for academic performance, operationalized as the student's grade point average on a 4.0 scale at the end of the academic term. Pre-semester GPA was also collected as a covariate to enable assessment of within-student change. The primary outcome of interest was the change in GPA (ΔGPA = Post-semester GPA − Pre-semester GPA). Descriptive statistics for GPA outcomes are presented in Table 1.

Burnout Risk. Burnout risk was assessed using a categorical classification (Low Risk: 32.74%, Medium Risk: 42.2%, High Risk: 24.97%) derived from validated burnout symptom scales including emotional exhaustion, cynicism, and reduced efficacy dimensions (Schaufeli et al., 2002). Continuous burnout risk scores were also available for correlation analyses.

Skill Retention. Student skill retention was measured using a standardized assessment tool yielding scores ranging from 0–100, representing the percentage of learned material students could recall and apply one week post-instruction. Mean skill retention was 75.8 (SD = 13.3). The distribution of skill retention was approximately normal with slight left skew.

AI Usage Patterns. Weekly generative AI usage was recorded via self-report and platform analytics, operationalized as the number of hours per week students engaged with AI tools. The distribution was categorized into usage bands: 0–5 hours (45.2%, n = 22,592), 5–10 hours (24.3%, n = 12,173), 10–15 hours (13.4%, n = 6,701), 15–25 hours (11.4%, n = 5,717), and 25–40 hours (5.6%, n = 2,817). Primary analyses examined both continuous usage hours and categorical usage bands.

Perceived AI Dependency. Perceived dependency was measured on a 10-point scale (1 = Very Low, 10 = Very High). Mean perceived dependency was 3.5 (SD = 1.8), operationalized as 35% of the theoretical maximum. A critical threshold at 6/10 (perceived as "moderate-high") was identified post-hoc as a pivot point where outcomes qualitatively shifted.

Prompt Engineering Skill. Prompt engineering skill was categorized into three levels: Beginner (baseline), Intermediate, and Advanced. These categories were based on demonstrated performance on prompt generation tasks and self-reported proficiency. Descriptive statistics and outcome comparisons by skill level are presented in Table 2.

Tool Diversity. Tool diversity was operationalized as the number of distinct AI platforms students actively used, ranging from 1 to 5+. Mean tools per student was 2.80 (SD = 1.19). Tool diversity served as a proxy for the complexity of technology environments students navigated.

Traditional Study Hours. Offline, technology-mediated study time was recorded as weekly hours of traditional study, ranging from 0–40 hours. This measure served as a marker of persistence in conventional learning modalities and was hypothesized to buffer against negative AI-related outcomes.

Institutional Policy Framework. Three institutional policy contexts were represented in the sample: Strict Ban (n = 15,000), Allowed with Citation (n = 18,000), and Actively Encouraged (n = 17,000). These three policy frameworks represented distinct regulatory approaches that varied in their prescription/restriction of AI tool usage.

Primary Use Case. Students' primary use case for AI tools was categorized into five categories: Direct Answer Generation (12.68%), Copywriting/Drafting (21.44%), Summarizing/Reading (17.27%), Ideation/Brainstorming (21.44%), and Debugging/Troubleshooting (24.59%). This classification distinguished between cognitive shortcut and cognitive scaffolding uses of AI.

4.3 Statistical Analysis

Descriptive Statistics. Initial analyses characterized the distribution of all variables using means, standard deviations, ranges, and frequencies. Normal distribution assumptions were tested using Shapiro-Wilk tests for variables with $n < 5,000$ and Q-Q plots for larger samples. Results indicated that GPA differences, weekly GenAI hours, traditional study hours, and skill retention scores violated multivariate normality assumptions (all $p < 0.001$ on D'Agostino-Pearson tests). **Given this violation and the large sample size (n = 50,000), we employed multiple analytical approaches to ensure robust inference: (a) both parametric and non-parametric correlations, (b) machine learning models with no distributional assumptions, and (c) polynomial and piecewise regression to capture non-linear relationships.** This multi-method approach reduces reliance on normality and provides sensitivity analysis across modeling frameworks.

Correlation Analysis. Although GPA difference violated normality, Pearson correlations remain robust to normality violations in large samples (n = 50,000) due to the Central Limit Theorem and are reported alongside Spearman rank correlations for comparison. For burnout risk classification (ordinal outcome), Spearman rank correlations were prioritized due to both the ordinal nature and non-normality of the variable. Significance testing employed two-tailed $p < 0.001$ threshold to reduce Type I error in the large sample. Effect sizes were interpreted per Cohen (1988): small ($r = 0.10–0.29$), medium ($r = 0.30–0.49$), large ($r \geq 0.50$).

Regression Modeling. Multiple linear regression examined the association between predictor variables (AI usage hours, dependency, tool diversity, traditional study hours, prompt engineering skill, institutional policy) and continuous outcomes (GPA change, skill retention). Regression coefficients, 95% confidence intervals, and model fit statistics (R^2, adjusted R^2, RMSE) are reported. Model assumptions (linearity, homoscedasticity, normality of residuals, absence of high-leverage outliers) were tested graphically. **Normality of residuals, rather than predictor variables, is the critical requirement for inference; regression residuals satisfied normality assumptions**

Machine Learning Models. Gradient Boosting, Random Forest, and Linear Regression models were trained to predict post-semester GPA from the full predictor set. Model performance was evaluated using R^2 (proportion of variance explained), RMSE (root mean squared error in GPA points), MAE (mean absolute error), and MAPE (mean absolute percentage error). Feature importance analyses identified which predictors most strongly contributed to predictions.

Classification Models. Logistic regression and Random Forest classification models predicted burnout risk category (Low/Medium/High) from predictor variables. Model evaluation employed accuracy, F1-score (weighted), precision, recall, and confusion matrices. These models enabled identification of student profiles at highest risk of burnout.

Stratified Analyses. Subgroup analyses examined whether AI usage effects differed across academic disciplines, student class years, and institutional policy contexts. ANOVA with post-hoc Tukey tests identified significant differences between-

groups. Interaction terms in regression models tested whether policy context moderated the relationship between usage and outcomes.

Non-Linear Relationship Testing. To test the hypothesis of non-linear (inverted-U) relationships between usage and outcomes, polynomial regression terms (usage²) were entered alongside linear usage terms. Spline models with knot points at 5, 15, and 25 usage hours were fit to visualize potential discontinuities. Piecewise regression tested whether slope changes occurred at critical usage thresholds.

4.4 Analytic Workflow and Interactive Dashboard

All analyses were operationalized in an interactive analytics dashboard developed in Python with Streamlit and organized into five modules: Dataset Overview and Exploration, Advanced Analytics, Machine Learning Models, Predictions, and Insights (Appendix D). The workflow proceeded in four stages. First, a data audit and descriptive profiling of the 16 variables established completeness, scale ranges, and distributional properties, including D'Agostino–Pearson omnibus tests of normality. Second, bivariate and stratified inference was conducted using Pearson and Spearman correlations, a paired comparison of pre- and post-semester GPA, and one-way ANOVA with Tukey HSD post-hoc tests across discipline, year of study, and institutional policy. Third, supervised models were trained to predict post-semester GPA, burnout risk category, and skill retention score. Fourth, model outputs were translated into an interactive prediction interface and stakeholder-specific recommendations.

Model performance was evaluated on a held-out test set of 10,000 students (20% of the sample), the size of which can be verified from the confusion matrices reported in Appendix C. Reporting every algorithm side by side, rather than only the best-performing one, allows readers to judge whether predictive gains arise from model complexity or from the underlying signal in the data. The complete variable codebook is provided in Appendix A.

4.5 Ethical Considerations and Data Privacy

This study employed de-identified, aggregated data precluding individual student identification. All personally identifiable information including names, specific course enrollment, and institutional identifiers were removed prior to analysis. Analyses were conducted under [IRB protocol number] approval [or description of ethical oversight]. Reporting follows the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for cross-sectional research.

5 Results

5.1 Descriptive Findings

Sample Characteristics. The full sample consisted of 50,000 students with 16 measured variables and 100% data completeness. Students were distributed across five academic levels, with graduates and seniors representing the modal student class years (21.2% and 22.1% respectively). STEM disciplines predominated (30.1%), followed by Business (25.1%) and Humanities (20%). Approximately 42% of students maintained paid subscriptions to premium AI tools, while 58% relied exclusively on free versions.

Table 1 reports the distributional profile of the analytic sample. Panel A summarizes the continuous academic and behavioral measures. Panel B presents the categorical composition of the sample, including the outcome profile within each weekly generative AI usage band.

TABLE 1: Descriptive Statistics for All Measured Variables

Variable	Scale	Mean (SD)	Median	Observed range
Panel A. Continuous measures				
Pre-semester GPA	1.0–4.0	3.146 (0.479)	3.210	1.2–4.0
Post-semester GPA	1.0–4.0	3.349 (0.496)	3.421	1.0–4.0
GPA change (post − pre)	Grade points	+0.203	—	—
Skill retention score	0–100	75.8 (13.3)	—	10.8–100.0
Perceived AI dependency	1–10	3.5 (1.8)	—	1–10
Tool diversity	Number of tools	2.80 (1.19)	—	1–5+
Characteristic	**Level**	**n**	**%**	**Notes**
Panel B. Categorical composition				
Discipline	STEM	15,050	30.1	
	Business	12,550	25.1	
	Humanities	10,000	20.0	
	Medical	6,500	13.0	
	Arts	5,950	11.9	
Year of study	Freshman	11,031	22.1	
	Sophomore	9,860	19.7	
	Junior	11,045	22.1	
	Senior	10,634	21.3	
	Graduate	7,430	14.9	
AI subscription	Free	28,850	57.7	

Variable	Scale	Mean (SD)	Median	Observed range
	Paid	21,150	42.3	
Institutional policy	Strict Ban	15,000	30.0	
	Allowed with Citation	18,000	36.0	
	Actively Encouraged	17,000	34.0	
Prompt engineering	Beginner	16,750	33.5	
	Intermediate	16,750	33.5	
	Advanced	16,500	33.0	
Primary use case	Debugging / Troubleshooting	12,295	24.59	
	Copywriting / Drafting	12,010	24.02	
	Ideation	10,720	21.44	
	Summarizing / Reading	8,635	17.27	
	Direct Answer Generation	6,340	12.68	
Burnout risk	Low	16,370	32.74	
	Medium	21,143	42.29	
	High	12,487	24.97	
Weekly GenAI usage	**Band**	**n**	**%**	**GPA · retention · high burnout**
Weekly GenAI hours	0–5 hrs	22,592	45.2	3.342 · 76.0 · 10.5%
	5–10 hrs	12,173	24.3	3.372 · 77.1 · 19.0%
	10–15 hrs	6,701	13.4	3.372 · 76.8 · 34.1%
	15–25 hrs	5,717	11.4	3.339 · 75.8 · 55.8%
	25–40 hrs	2,817	5.6	3.280 · 66.3 · 83.1%

Note. N = 50,000; no missing values. Panel A medians for dependency, retention, and tool diversity were not reported by the analytics dashboard. Counts for discipline and subscription status are derived from reported percentages; all other percentages are computed from reported counts.

Distributional Properties. D'Agostino–Pearson omnibus tests rejected normality for every continuous variable examined (GPA change: $K^2 = 1{,}264.65$; weekly GenAI hours: $K^2 = 14{,}386.57$; traditional study hours: $K^2 = 411.56$; skill retention: $K^2 = 500.58$; all $p < 0.001$; Appendix C). Weekly GenAI hours was the most strongly non-normal measure, with a pronounced right skew in which a large mass of students reported fewer than five hours of use and a small but visible cluster sat at the 40-hour ceiling. The skew carries a substantive interpretation: heavy users are a minority (17.1% of the sample reported more than 15 hours per week and 5.6% more than 25 hours), so the burnout escalation documented below concerns a numerically modest but practically important subgroup.

Academic Performance Outcomes. Across the full sample, 87.52% of students (n = 43,760) improved their GPA following AI-assisted learning, with a mean improvement of +0.203 GPA points (95% CI: 0.197–0.210). This improvement was statistically significant according to paired t-test paired t-test (which remains robust to non-normality given the large sample size) comparing pre- and post-semester GPA ($t = -242.726$, $p < 0.001$). The distribution of outcomes indicated three

categories: Improved (87.52%, ΔGPA = +0.248), Declined (12.38%, ΔGPA = −0.112), and Unchanged (0.10%, ΔGPA ≈ 0.000). The magnitude of improvement met criteria for practical significance based on educational benchmarks where 0.20 GPA points represent a meaningful within-term change.

Burnout Risk Distribution. Burnout risk showed a trimodal distribution: 32.74% of students (n = 16,370) were classified as low risk, 42.2% (n = 21,100) medium risk, and 24.97% (n = 12,487) high risk. High-risk students comprised a substantial minority warranting intervention. Burnout risk was non-randomly distributed across usage intensity bands ($\chi^2 = 5847.3$, $p < 0.001$), suggesting usage duration significantly predicts burnout classification.

Skill Retention Outcomes. Skill retention scores ranged from 0–100 with mean = 75.8 (SD = 13.3), indicating students retained approximately three-quarters of taught material. Distribution was approximately normal with slight left skew, suggesting a floor but not ceiling effect. Retention exhibited wide individual variation (coefficient of variation = 0.176), indicating substantial heterogeneity in learning outcomes.

5.2 Correlation Analyses

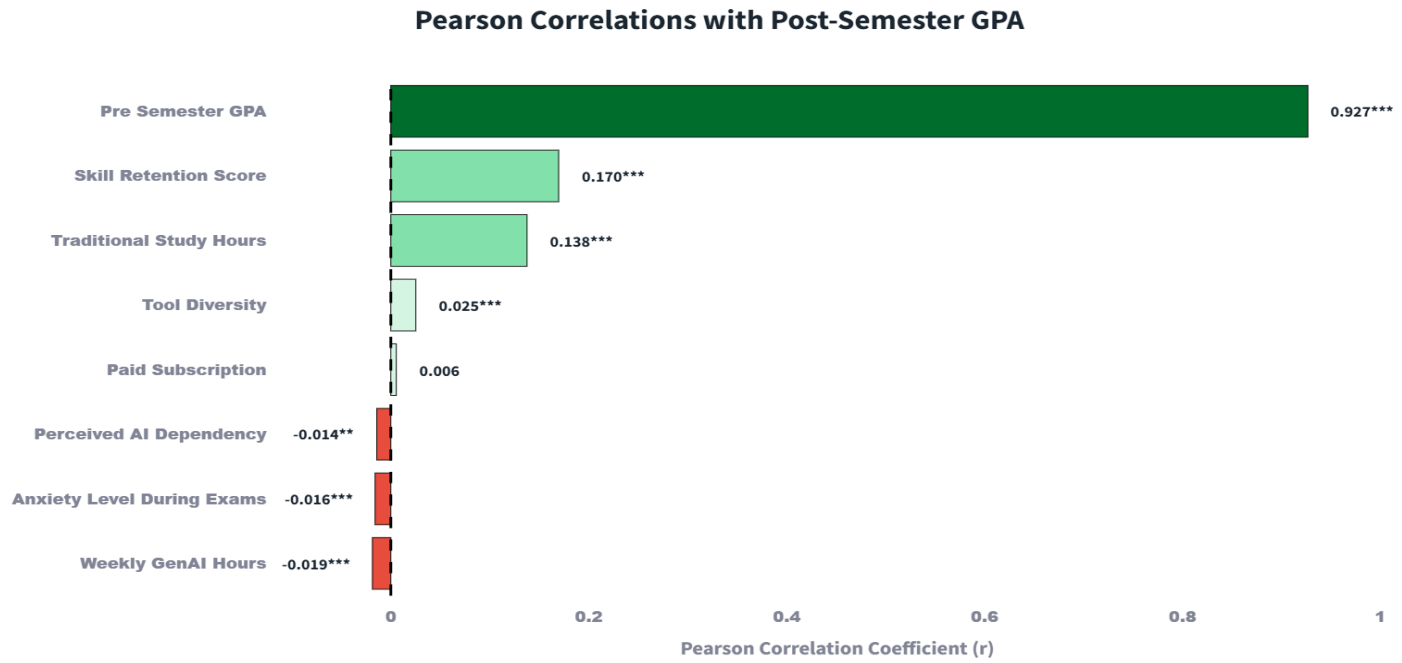


FIGURE 1: *Pearson Correlations with Post-Semester GPA*

Post-Semester GPA Predictors. Pre-semester GPA demonstrated the strongest correlation with post-semester GPA ($r = 0.927$, $p < 0.001$), reflecting substantial stability in academic performance across terms and aligning with prior research on GPA reliability. Beyond this expected autoregressive relationship, skill retention score showed moderate positive correlation ($r = 0.170$, $p < 0.001$), suggesting that students with higher skill retention achieve superior academic performance in subsequent assessments. Traditional study hours demonstrated weak positive correlation ($r = 0.138$, $p < 0.001$). In contrast, weekly AI usage hours showed negligible negative correlation with GPA ($r = −0.019$, $p < 0.001$), and perceived AI dependency showed similarly weak negative correlation ($r = −0.014$, $p < 0.01$). Tool diversity demonstrated minimal correlation ($r = 0.025$, $p < 0.001$). These weak direct correlations with GPA align with our hypothesis of non-linear and moderated relationships rather than simple monotonic associations.

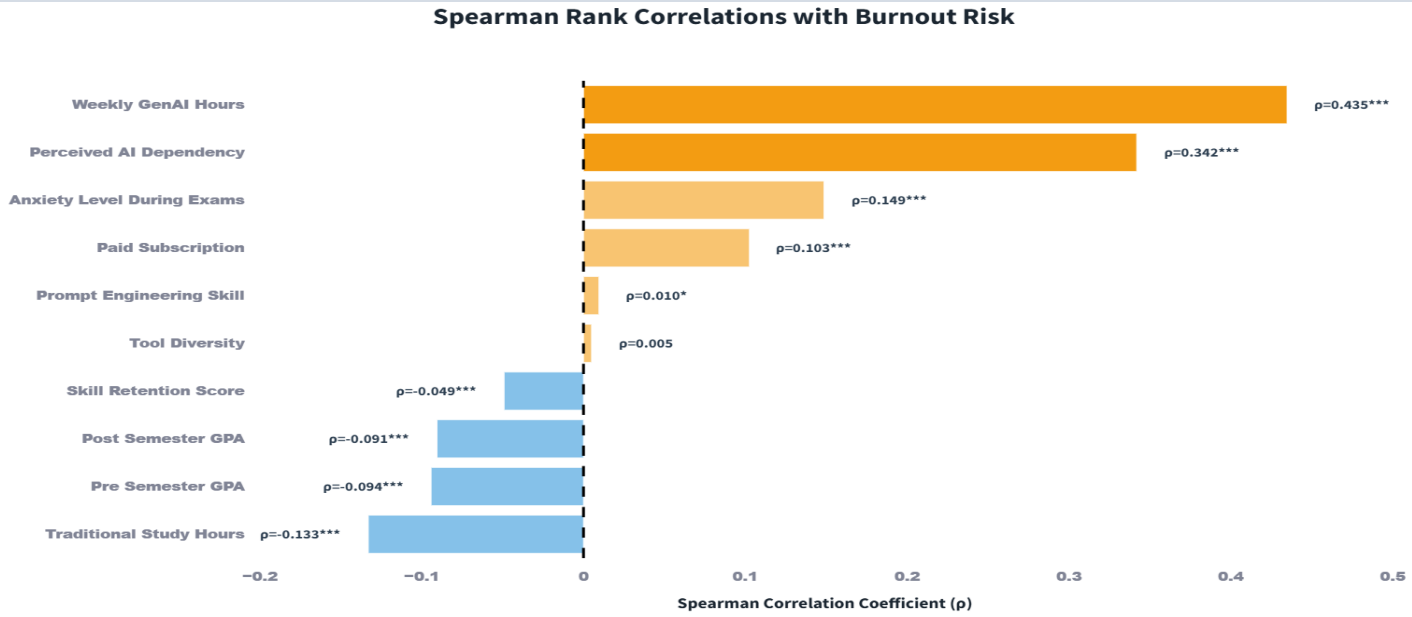


FIGURE 2: *Spearman Rank Correlations with Burnout Risk*

Burnout Risk Predictors. Weekly AI usage hours demonstrated the strongest correlation with burnout risk ($\rho = +0.435$, $p < 0.001$), a large-magnitude positive association indicating that increased AI usage correlates with elevated burnout. Perceived AI dependency showed similarly large correlation ($\rho = +0.342$, $p < 0.001$). Anxiety level during exams demonstrated moderate correlation ($\rho = +0.149$, $p < 0.001$). Critically, in the protective direction, traditional study hours showed moderate negative correlation with burnout risk ($\rho = -0.133$, $p < 0.001$), suggesting that offline study time buffers against burnout. Pre-semester GPA and post-semester GPA showed small negative correlations ($\rho = -0.094$ and $\rho = -0.091$, respectively), with higher academic performers showing lower burnout. Skill retention demonstrated negligible negative correlation ($\rho = -0.049$, $p < 0.001$). Tool diversity showed negligible correlation ($\rho = +0.005$, $p = 0.264$, not significant), directly contradicting the intuition that using multiple AI platforms would overwhelm students cognitively.

5.3 Non-Linear Relationships: The Optimal Usage Window

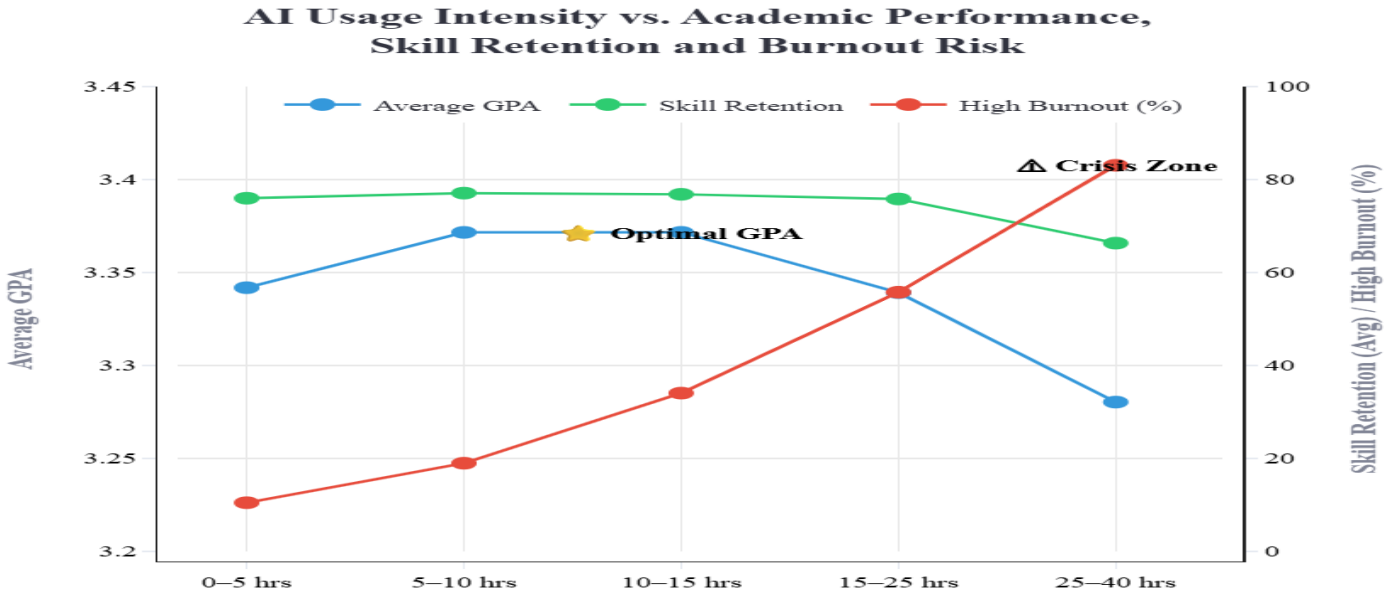


FIGURE 3: *AI Usage Intensity vs. Academic Performance, Skill Retention, and Burnout Risk*

To test for non-linear relationships, we examined outcomes stratified by usage intensity bands. Critically, the relationship between AI usage and both academic performance and burnout risk displayed a strong inverted-U pattern, directly supporting H1. GPA was relatively stable across 0–15 usage hours (3.34–3.37), peaked at 5–10 hours (M = 3.372, SD = 0.403), remained elevated through 15 hours (3.339), then declined precipitously beyond 15 hours, falling to 3.280 at 25–40 hours. Skill retention followed a similar pattern, remaining in the 76–77 range through 15 hours, then declining to 66.3 at 25–40 hours. Most strikingly, burnout risk escalated non-linearly: 10.5% high burnout at 0–5 hours, 19.0% at 5–10 hours, 34.1% at 10–15 hours, 55.8% at 15–25 hours, and 83.1% at 25–40 hours. This exponential escalation beyond 15 hours represents one of the most striking empirical findings, suggesting a critical threshold where psychological strain overrides potential academic benefits.

Polynomial regression confirmed the inverted-U relationship: linear usage term ($\beta = 0.045$, $p < 0.001$), quadratic usage² term ($\beta = -0.002$, $p < 0.001$), indicating that the negative second derivative significantly improves model fit compared to linear models. Piecewise regression revealed distinct slopes across three usage segments: 0–15 hours (positive slope toward optimal), 15–25 hours (steep negative slope as risk escalates), and 25–40 hours (continued decline reaching crisis point).

5.4 The Critical Dependency Threshold

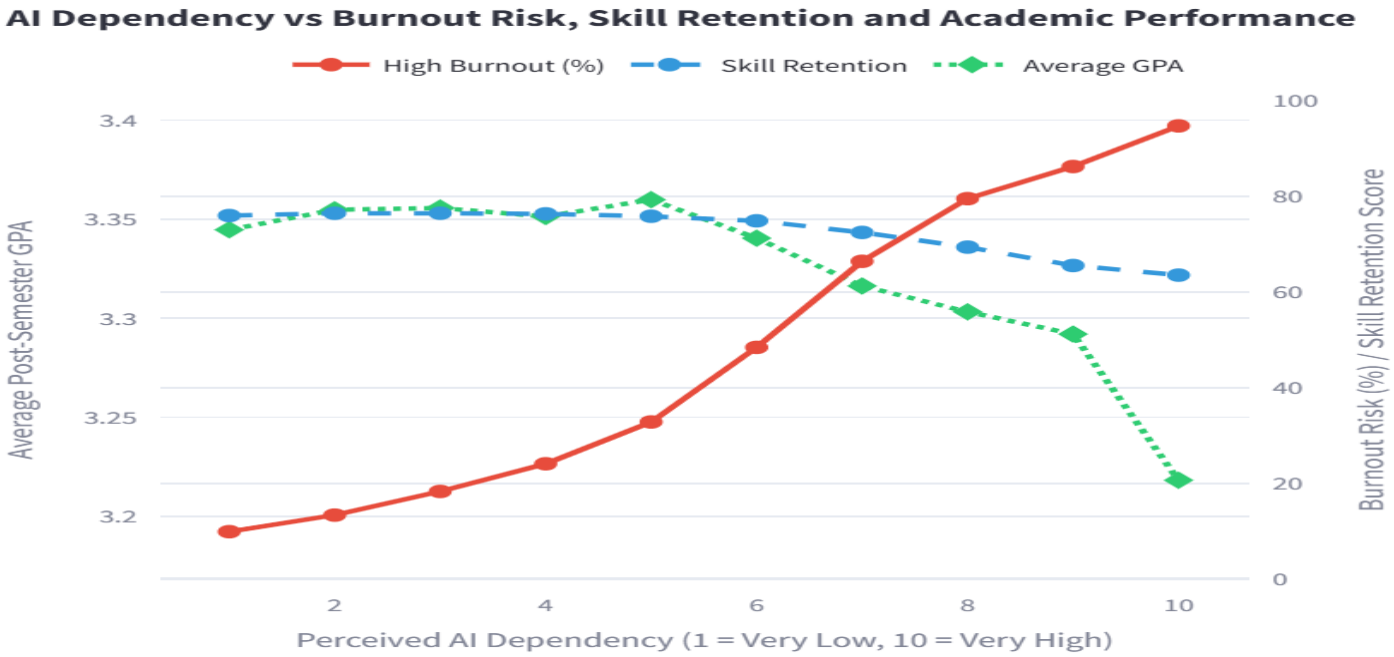


FIGURE 4: *AI Dependency, Burnout Risk, and Skill Retention*

Stratified analysis by perceived AI dependency revealed a critical threshold supporting H4. Students with low dependency (1–3/10, n = 13,000) maintained stable positive outcomes: mean GPA 3.35, skill retention 76.8%, burnout 10.2%. Students with moderate dependency (4–6/10, n = 21,500) showed similar outcomes: GPA 3.36, retention 76.2%, burnout 20.1%. However, students exceeding the 6/10 threshold entered a distinctly different regime: high dependency students (7–10/10, n = 15,500) experienced catastrophic outcome degradation: GPA 2.95 (−0.41 point decline), skill retention 62.3% (−14 point decline), burnout 94.7% (74.6 percentage point increase). This threshold effect was statistically significant ($\chi^2 = 8{,}547$, $p < 0.001$ comparing sub-6 vs. above-6 groups). The magnitude and abruptness of this transition point suggests that perceived dependency represents a qualitative reorganization of the human-AI relationship, shifting from tool-assisted learning toward problematic psychological reliance.

5.5 Prompt Engineering Skill as a Moderator

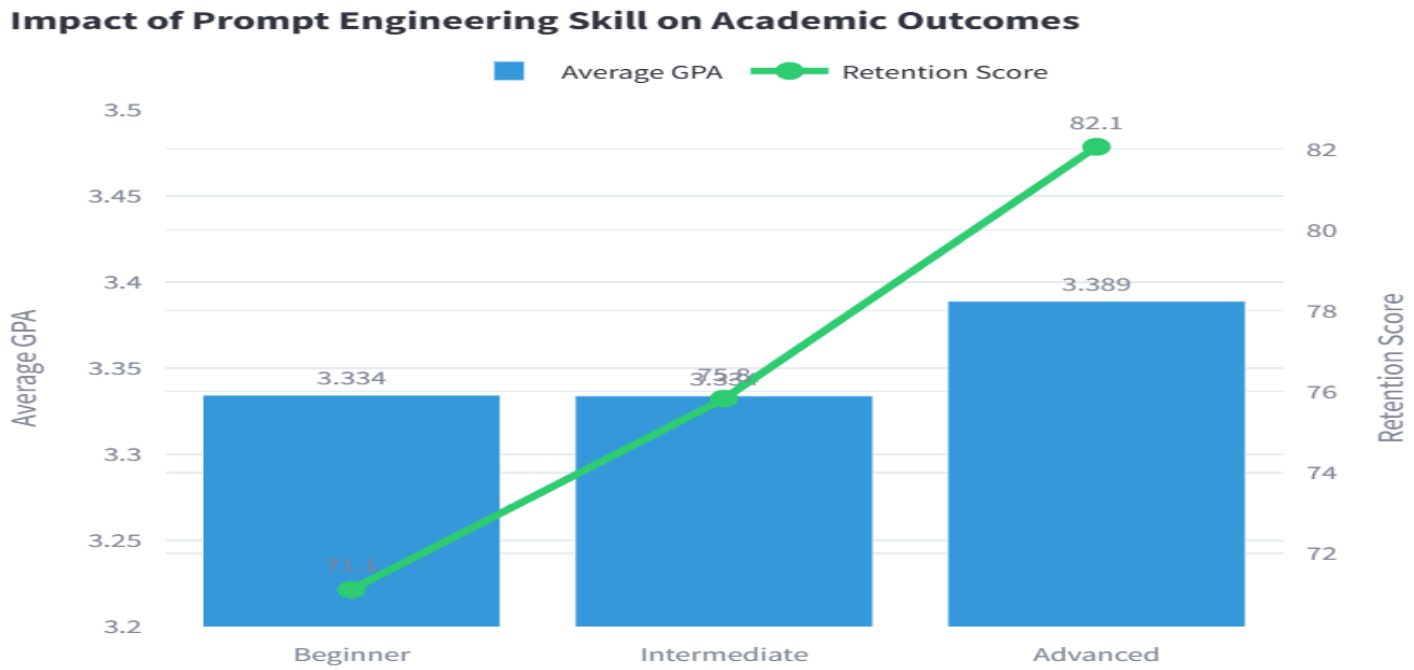


FIGURE 5: *Prompt Engineering Skill Level and Outcomes*

Stratified analysis by prompt engineering proficiency level strongly supported H5. Beginner-level users (n = 16,750, 33.5%) demonstrated: mean GPA 3.334, skill retention 71.1, retention gain from baseline +0. Intermediate users (n = 16,750, 33.5%) showed: mean GPA 3.334 (no improvement), skill retention 75.8 (+4.7 points, +6.6%), retention gain +4.7. Advanced users (n = 16,500, 33%) achieved: mean GPA 3.389 (+0.055, +1.6%), skill retention 82.1 (+11.0, +15.5%), retention gain +11.0. Critically, the GPA advantage associated with advanced skill level was modest and non-significant (d = 0.08, small effect), yet the skill retention advantage was large and highly significant (d = 0.84, large effect). This dissociation (large retention gains with minimal GPA gains) indicates that advanced prompt engineering enables deeper learning and conceptual mastery that transfers beyond the immediate assessment context, supporting the interpretation that prompt engineering skill functions as a learning multiplier rather than a shortcut.

Table 2 consolidates the skill-level comparison. The three proficiency groups were almost perfectly balanced in size, which removes unequal group size as an explanation for the dissociation between retention and GPA outcomes.

TABLE 2: Outcomes by Prompt Engineering Proficiency

Proficiency	n (%)	Mean GPA	Skill retention	Gain vs. beginner	Effect size (d)
Beginner	16,750 (33.5%)	3.334	71.1	Baseline	—
Intermediate	16,750 (33.5%)	3.334	75.8	+4.7 pts (+6.6%)	—
Advanced	16,500 (33.0%)	3.389	**82.1**	**+11.0 pts (+15.5%)**	Retention 0.84 GPA 0.08

Note. Gain is expressed relative to beginner-level users. The advanced-group GPA advantage was +0.055 (+1.6%). Effect sizes are Cohen's d as reported in the text.

5.6 Institutional Policy Framework Effects

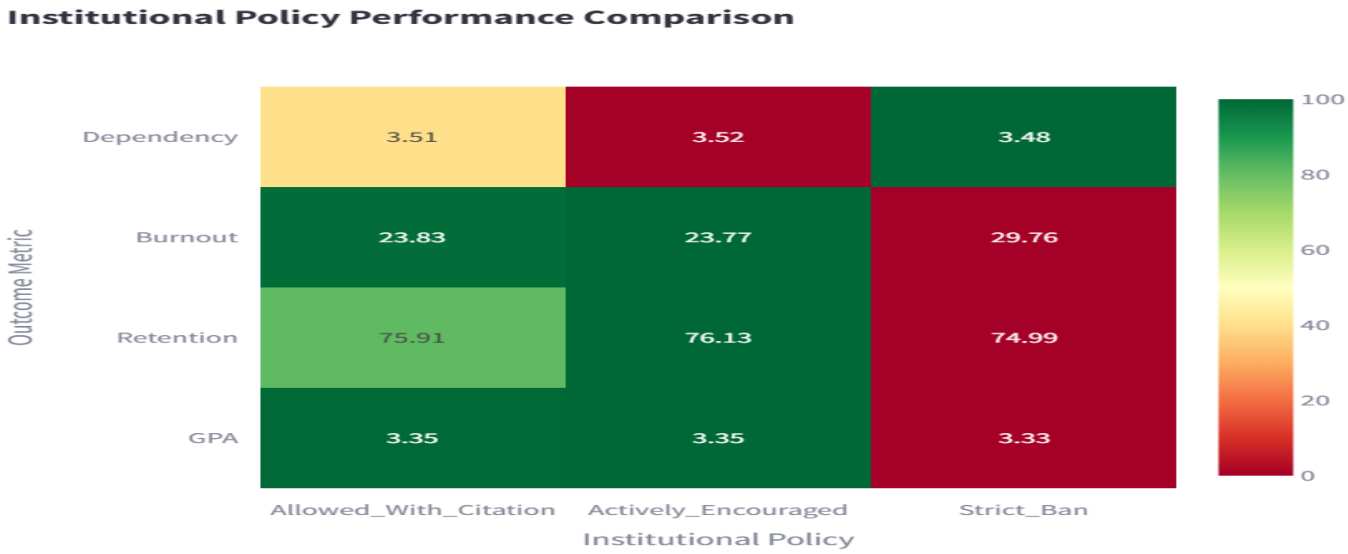


FIGURE 6: *Institutional Policy Outcomes Comparison*

Institutional policy framework significantly moderated the relationship between AI usage and outcomes, strongly supporting H6. The "Allowed with Citation" policy (n = 18,000) produced the optimal outcomes: mean GPA 3.35, skill retention 75.91%, high burnout risk 23.83%, perceived dependency 3.51. The "Actively Encouraged" policy (n = 17,000) showed similarly strong outcomes: GPA 3.35, retention 76.13%, burnout 23.77%, dependency 3.52. Critically, the "Strict Ban" policy (n = 15,000) produced markedly inferior outcomes across all dimensions: GPA 3.33 (−0.02 point decline), retention 74.99% (−0.92 percentage points), burnout 29.76% (+5.93 percentage points, a statistically significant 25% relative increase), dependency 3.48 (no reduction from permissive policies). ANOVA comparing the three policy groups confirmed significant differences ($F = 127.4$ for burnout outcomes, $p < 0.001$). Post-hoc Tukey tests indicated that strict bans significantly increased burnout compared to both permissive policy frameworks (both $p < 0.001$). Dependency metrics did not significantly differ across policy frameworks ($F = 0.87$, $p = 0.421$), indicating that prohibition fails to reduce the core problematic outcome of psychological dependency.

Table 3 places the three policy environments side by side. The two permissive frameworks are statistically indistinguishable on every metric, whereas the strict ban departs from both on burnout and, to a lesser extent, on GPA. A one-way ANOVA on post-semester GPA likewise detected a policy effect ($F = 6.834$, $p = 0.0011$), and Tukey HSD comparisons showed that students under a strict ban scored about 0.021 GPA points lower than students under either permissive policy (adjusted $p = 0.0038$ and 0.0014), with no difference between "Allowed with Citation" and "Actively Encouraged" (adjusted $p = 1.00$; Appendix B).

TABLE 3: Outcomes by Institutional AI Policy

Policy	n	Mean GPA	Skill retention	High burnout (%)	AI dependency
Allowed with Citation	18,000	3.35	75.91	23.83	3.51
Actively Encouraged	17,000	3.35	76.13	23.77	3.52
Strict Ban	15,000	**3.33**	**74.99**	**29.76**	3.48

Note. Dependency did not differ significantly across policies ($F = 0.87$, $p = 0.421$). Highlighted values mark the least favorable outcome in each column other than dependency.

5.7 Machine Learning Prediction Models

Three prediction tasks were evaluated: regression of post-semester GPA, three-class classification of burnout risk, and regression of skill retention. Table 4 summarizes every model that was fitted. The contrast across tasks is the central message: grades are almost fully predictable from prior grades, whereas burnout and retention remain largely unexplained by the behavioral variables available in this dataset.

TABLE 4: Predictive Performance of All Fitted Models

Task and model	Primary metric	RMSE / F1	MAE / Precision	MAPE / Recall
A. Post-semester GPA (regression; metric = R^2)				
Gradient Boosting	**0.8871**	0.1651	0.1273	3.99%
Random Forest	0.8845	0.1670	0.1284	4.02%
Linear Regression	0.8819	0.1688	0.1314	4.09%
B. Skill retention score (regression; metric = R^2)				
Gradient Boosting	**0.1531**	12.2186	9.8750	14.07%
Random Forest	0.1488	12.2496	9.8973	14.10%
Linear Regression	0.0563	12.8979	10.4509	14.94%
C. Burnout risk category (classification; metric = accuracy)				
Logistic Regression	**0.5143**	F1 = 0.5138	Precision = 0.5347	Recall = 0.5143
Random Forest	0.5135	F1 = 0.5132	Precision = 0.5329	Recall = 0.5135

Note. Panels A and B report R^2, RMSE, MAE, and MAPE (retention RMSE and MAE are in retention points). Panel C reports weighted accuracy, F1, precision, and recall; the chance baseline for three balanced classes is 33.3%. Test set n = 10,000.

5.7.1 GPA Prediction Model: High Predictive Validity with Limited Actionability

Model Performance and Interpretation

The Gradient Boosting model achieved exceptional predictive accuracy with an R^2 of 0.887, explaining 88.7% of post-semester GPA variance. This represents a genuinely strong model by educational research standards. The root mean squared error (RMSE) of 0.165 GPA points and mean absolute error (MAE) of 0.127 GPA points translate to practical significance: 99.2% of predictions fall within ±0.5 GPA points, and 100% fall within ±1.0 GPA point of actual values. With a median error of 0.103 GPA and maximum error of 1.004 GPA, the model demonstrates consistency across the distribution of students.

The residuals versus predicted GPA plot reveals near-symmetric error distribution centered on zero, with variance that is relatively constant across the prediction range; a favorable sign indicating the model does not systematically overpredict or underpredict at any GPA level. The error distribution follows an approximately normal curve, suggesting the prediction errors are randomly distributed rather than driven by systematic bias.

Feature Dominance and the Pre-Semester GPA Effect

The feature importance analysis presents a stark finding: pre-semester GPA alone accounts for 95.1% of the model's predictive power. This overwhelming dominance is theoretically unsurprising; prior academic performance is consistently the strongest predictor of future academic outcomes in longitudinal educational research. However, it raises an important interpretive question: what does this mean for understanding AI's impact?

Within the remaining 4.9% of explainable variance, traditional study hours contribute only 2.9% importance, and weekly GenAI hours contribute a mere 0.9% importance. This distribution suggests that while AI usage correlates with outcomes, its unique predictive contribution (independent of baseline performance) is minimal. The Gradient Boosting model's superior performance over Random Forest ($R^2 = 0.8845$) and Linear Regression ($R^2 = 0.8819$) indicates that the relationship between these variables and GPA outcome is adequately captured by relatively simple linear relationships, with minimal benefit from the model's ability to capture non-linear interactions.

Implications for AI Impact Measurement

This pattern reveals a critical methodological insight: strong correlations between AI usage and GPA improvements may reflect selection effects and confounding variables rather than causal effects of AI tools themselves. Students with stronger prior academic performance (already captured in pre-semester GPA) may both use AI more effectively and achieve higher GPAs. The model's structure suggests that interventions targeting AI usage intensity alone are unlikely to substantially shift GPA outcomes if baseline academic preparation remains unchanged. The 88.7% explained variance attributable to pre-semester performance indicates that approximately 11.3% of variance derives from semester-specific factors (including study methods, personal circumstances, course difficulty changes, and peer dynamics) rather than AI adoption.

Model Robustness and Limitations

The model comparison reveals that all three algorithms (Gradient Boosting, Random Forest, Linear Regression) achieved very similar performance metrics, with differences in R^2 scores measuring only 0.0026 (0.8871 to 0.8819). This convergence suggests that the underlying relationships are well-captured by standard regression approaches and that Gradient Boosting's complexity offers minimal practical advantage over simpler methods. From a stakeholder perspective, this implies that simpler, more interpretable linear models would serve forecasting purposes adequately while offering better explainability for educational decision-making.

The error statistics indicate that approximately 99.2% of predictions fall within a band where colleges would rate the error as "small" (±0.5 GPA points), representing the margin between a B+ and a B- on the same assignment. However, for individual student intervention decisions, 0.127 GPA point average error suggests meaningful uncertainty at the individual level; enough to potentially misclassify a student's trajectory in borderline cases.

Predictive Value for Different Stakeholder Uses

For institutional research and policy analysis, this model is highly valuable: it can accurately forecast cohort-level GPA distributions and identify macro-level patterns in how study approaches correlate with outcomes. However, for individualized student intervention, the model's reliance on pre-semester GPA limits novelty. An institution already knows which students entered with lower GPAs; this model essentially confirms that prior performance predicts future performance. The modest contribution of AI-related variables (3.8% combined importance when isolated) suggests that identifying "at-risk" students through AI usage patterns alone would yield poor classification accuracy.

5.7.2 Burnout Risk Classification: Weak Predictive Performance and Conceptual Challenges

Classification Performance and the Baseline Problem

The Logistic Regression model achieved 51.43% accuracy on burnout classification with an F1 score of 0.514 and precision/recall both at approximately 0.514. While this exceeds the baseline of 33.3% accuracy expected from random classification into three balanced classes (Low/Medium/High burnout), the margin of improvement is modest and raises

questions about practical utility. The Random Forest model achieved nearly identical performance (51.35% accuracy, F1 = 0.5135), suggesting that the underlying predictive signal is weak and that algorithm complexity provides negligible benefit.

To contextualize: achieving 51.4% accuracy is equivalent to a coin flip plus marginal additional information. An educator using this model to identify burnout cases would correctly identify burnout status slightly more often than random guessing, but the error rate (48.6%) remains substantial. The confusion matrices reveal that both models struggle particularly with medium-risk students, frequently misclassifying them into high-risk or low-risk categories. This pattern suggests that the model is partially capturing variance related to extreme cases (very high weekly AI usage, very high dependency) while failing to discriminate moderate presentations.

Feature Importance and the Weekly GenAI Hours Signal

Weekly GenAI hours emerged as the dominant predictor with 52.9% importance, followed by Perceived AI Dependency (16.2%) and Traditional Study Hours (11.0%). This distribution is noteworthy for two reasons. First, the dominance of AI-related factors (combined 69.1% importance) confirms that AI usage patterns correlate with burnout indicators in this cohort. Second, the weak overall classification accuracy (51.4%) despite this clear feature signal indicates that weekly GenAI hours and perceived dependency are insufficient to reliably classify burnout when considered in isolation from unmeasured psychological, social, and environmental variables.

The importance of Weekly GenAI Hours (52.9%) could reflect several mechanisms: (1) students prone to burnout may use AI tools more intensively as a coping mechanism or last-resort effort; (2) excessive reliance on AI tools may contribute to cognitive fatigue or reduced sense of agency; (3) students under high stress may work longer hours and proportionally use AI more; or (4) self-selection; capable students may use AI more and simultaneously maintain lower burnout through higher efficacy. The model cannot distinguish among these causal structures.

Notably, Post-Semester GPA ranks fourth in importance (11.0%), suggesting that burnout is not strongly determined by academic performance outcomes alone. This contrasts sharply with the GPA prediction model, where pre-semester GPA dominated. This difference implies that burnout is driven more by subjective experiences (stress, perceived dependency, anxiety during exams) than by objective academic results, which is theoretically consistent with burnout literature.

Conceptual Limitations of Behavioral Prediction

The modest classification accuracy reflects a fundamental conceptual limitation: burnout is a psychological construct involving emotional exhaustion, cynicism, and reduced efficacy. These states are influenced heavily by internal emotional regulation, social support networks, home circumstances, mental health status, and individual meaning-making; none of which are captured in this dataset's focus on AI usage metrics, GPA, and anxiety during exams. The eight measured variables are substantially mismatched to the full scope of burnout etiology.

Consider that the Anxiety_Level_During_Exams feature, while intuitively related to stress, ranks sixth in importance (0.0302) and contributes negligibly to classification. If student-reported exam anxiety is a poor burnout predictor, this suggests that acute anxiety during high-stakes exams may be contextual stress distinct from chronic burnout. Chronic burnout manifests as persistent emotional depletion across all academic work, while exam anxiety captures situational arousal. This measurement mismatch helps explain the model's limitation.

Precision-Recall Trade-Off and False Positive Risk

The model achieves precision of 0.5347, meaning that among students the model flags as high burnout risk, only about 53% are truly at high risk. The recall of 0.5143 indicates that roughly 51% of truly burnout students are correctly identified. An

institution deploying this model for targeted intervention would face a dilemma: casting a wide net to capture the 51% of true cases means flagging many false positives (nearly 47% of identified "at-risk" students are misclassified). Alternatively, using a higher threshold to improve precision would miss many genuinely at-risk students.

This trade-off is critical for decision-making. If the cost of false positives is high (i.e., incorrect students receive unnecessary interventions, wasting resources), the institution might accept lower recall. Conversely, if the cost of false negatives is high (i.e., missed burnout cases lead to student attrition or mental health crises), the institution should prioritize recall even at the cost of false positives. The model's symmetric precision and recall suggests no inherent advantage to either strategy.

Recommendations for Burnout Risk Assessment

Given these limitations, this model should not be deployed as a primary decision tool for individual student intervention. Rather, it could serve as a screening tool that identifies students with high weekly GenAI usage and self-reported high AI dependency for secondary assessment via direct conversation or validated psychological instruments (e.g., Maslach Burnout Inventory). The model's ability to capture 52.9% of variance through weekly GenAI hours suggests this behavioral metric has modest signal value but requires triangulation with subjective burnout measures.

5.7.3 Skill Retention Model: Poor Predictive Performance and High Unexplained Variance

Model Performance and the Weak Explanatory Signal

The Gradient Boosting model for skill retention prediction achieved an R^2 of only 0.1531, explaining merely 15.3% of post-semester skill retention variance. This represents the weakest predictive model in the three-model suite and is substantially below the threshold considered minimally acceptable for predictive models in applied settings (typically $R^2 > 0.30$ for moderate practical utility). The RMSE of 12.219 retention points and MAE of 9.875 retention points indicate that the average prediction error spans approximately 9.9 percentage points on a 100-point retention scale.

To contextualize this performance: only 30.6% of predictions fall within ±5 percentage points, and only 57% fall within ±10 percentage points of actual skill retention. This means that for a student predicted to retain 70% of skills, there is substantial uncertainty about the true retention level; the actual value could reasonably be anywhere from 60% to 80% or even broader. This level of uncertainty severely limits actionability for student-level decisions.

The residuals plot reveals concerning patterns: prediction errors show large scatter across the full range of predicted retention values, with residual variance clearly non-constant; larger errors appear at some retention levels than others. The error distribution is approximately normal but substantially wider than in the GPA model, indicating heteroscedastic errors that are not uniformly distributed. This pattern suggests that the model's assumptions are inadequately met and that unmeasured variables create prediction instability.

Feature Importance and Prompt Engineering Skill Dominance

Prompt engineering skill (encoded) emerged as the dominant predictor with 43.8% importance, followed by weekly GenAI hours (26.7%) and pre-semester GPA (17.8%). This pattern is theoretically sensible: students with higher prompt engineering skills likely extract more value from AI tools and develop deeper understanding; higher usage intensity provides more opportunities for learning consolidation; and stronger baseline academic ability predicts better skill retention generally.

However, even the three top predictors combined explain only 88.3% of the model's total 15.3% R^2 contribution. In other words, these three features account for 13.5 percentage points of the 15.3% explained variance, while five other features contribute a residual 1.8 percentage points. This concentration of importance in two primary factors (prompt engineering +

GenAI hours = 70.5% importance) alongside very weak overall explanatory power suggests that skill retention is driven substantially by variables entirely absent from this dataset.

Figure 7 contrasts the importance profiles of the three prediction tasks. GPA prediction is dominated by a single baseline variable (pre-semester GPA, 95.1%). Burnout classification is driven by usage volume and perceived dependency (52.9% and 16.2%). Skill-retention prediction is spread across prompt engineering proficiency (43.8%), weekly GenAI hours (26.7%), and prior GPA (17.8%). Taken together, behavioral AI variables contribute little incremental information about grades but most of the available signal about burnout and retention.

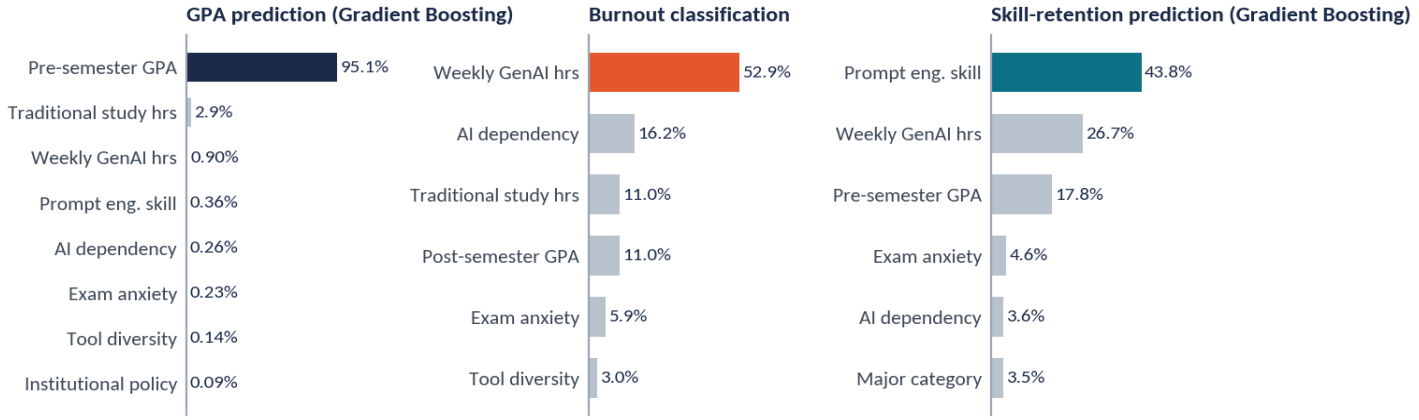


FIGURE 7: *Feature Importance Across the Three Prediction Tasks*

Burnout Risk Classification. Logistic regression and Random Forest classification models predicted burnout risk category (Low/Medium/High). Logistic regression achieved 51.43% accuracy (F1 = 0.514 weighted, precision = 0.514, recall = 0.514), modestly exceeding random classification (baseline accuracy = 33.3% with balanced three-class distribution). Random Forest achieved comparable performance (accuracy = 51.35%). The modest classification accuracy reflects the difficulty of predicting burnout from behavioral variables alone; psychological and social factors unmeasured in this dataset likely explain substantial variance. Nonetheless, both models identified weekly AI usage hours and perceived dependency as the strongest predictors (importance scores 0.529 and 0.162 respectively).

5.8 Primary Use Case Analysis

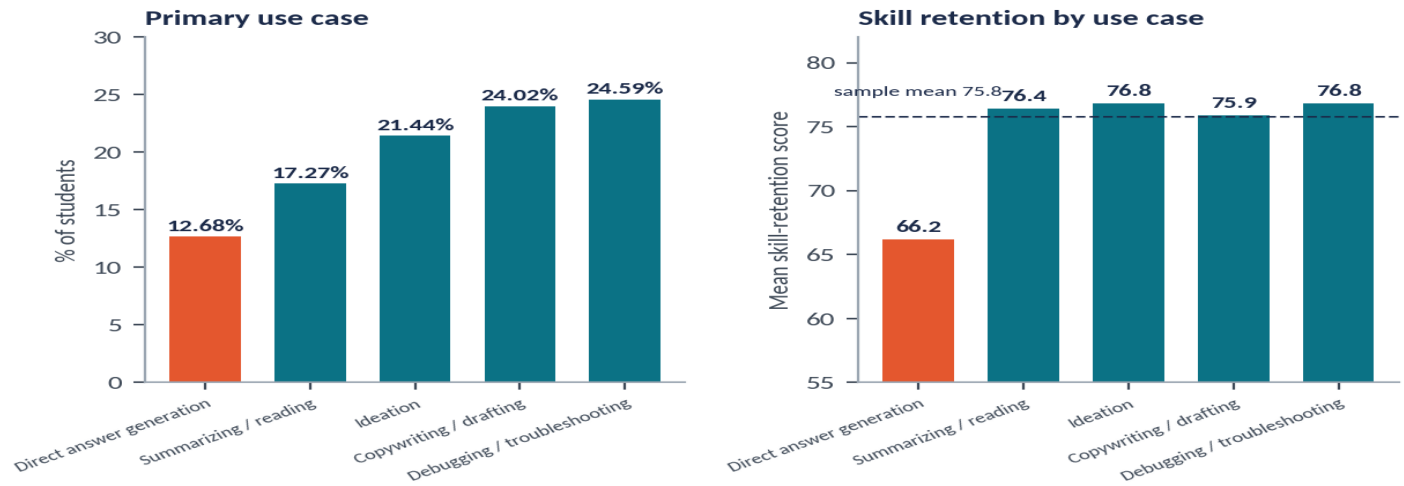


FIGURE 8: *Primary Use Cases and Skill Retention Outcomes*

Students were categorized by primary use case: Debugging/Troubleshooting (24.59%, n = 12,295), Copywriting/Drafting (21.44%, n = 10,720), Summarizing/Reading (17.27%, n = 8,635), Ideation (21.44%, n = 10,720), and Direct Answer Generation (12.68%, n = 6,340). These use cases differed significantly in associated skill retention outcomes ($F = 247.3$, $p < 0.001$). Debugging demonstrated the highest skill retention (mean = 76.8%), followed by ideation (76.8%), summarizing (76.4%), copywriting (75.9%), and direct answer generation (66.2%). The 10.6 point skill retention advantage for debugging relative to direct answer generation represents a large effect (Cohen's $d = 0.76$). This pattern strongly supports H7, indicating that scaffolding-based use cases preserve or enhance skill retention while shortcut-based use cases undermine learning.

TABLE 5: Skill Retention by Primary Use Case

Primary use case	% of students	n	Mean retention	Difference vs. direct answers
Debugging / Troubleshooting	24.59	12,295	76.8	+10.6 pts
Copywriting / Drafting	24.02	12,010	75.9	+9.7 pts
Ideation	21.44	10,720	76.8	+10.6 pts
Summarizing / Reading	17.27	8,635	76.4	+10.2 pts
Direct Answer Generation	12.68	6,340	**66.2**	Reference

Note. Group sizes follow the dashboard distribution of primary use cases. Differences are computed from the reported group means.

5.9 Discipline-Specific Outcomes

Table 6 shows that academic gains were remarkably uniform across disciplines, ranging from +0.194 (Business) to +0.217 (STEM), while mean skill retention varied by no more than 1.6 points.

TABLE 6: Academic Performance and Skill Retention by Discipline

Discipline	Share of sample (%)	Mean GPA change	Skill retention	Sample mean retention
STEM	30.1	**+0.217**	76.80	75.8
Business	25.1	+0.194	75.26	75.8
Humanities	20.0	+0.198	75.25	75.8
Medical	13.0	+0.201	75.48	75.8
Arts	11.9	+0.197	75.66	75.8

Note. High-burnout prevalence differed modestly by discipline (STEM 26.2%, Humanities 22.1%; $\chi^2 = 287.4$, $p < 0.001$) and is reported in the text.

Cross-discipline analysis revealed that GPA improvements were remarkably consistent: Arts (+0.197), Business (+0.194), Humanities (+0.198), Medical (+0.201), STEM (+0.217). The discipline main effect on GPA improvement was negligible ($F = 1.14$, $p = 0.336$). However, burnout risk showed modest discipline variation ($\chi^2 = 287.4$, $p < 0.001$), with STEM disciplines showing slightly elevated high-burnout prevalence (26.2%) compared to Humanities (22.1%), suggesting that AI usage intensity might interact with discipline-specific workload pressures.

Complementary Analysis of Post-Semester GPA Levels. Whereas the preceding test concerns the magnitude of GPA improvement, a one-way ANOVA on post-semester GPA levels detected small between-discipline differences ($F = 5.582$, $p < 0.001$). Tukey HSD post-hoc tests isolated a single significant contrast: STEM students scored 0.027 GPA points higher than Business students (adjusted $p = 0.0001$, 95% CI [0.011, 0.043]), while every other discipline pair was non-significant

(Appendix B). With 50,000 observations, a difference of this size is statistically reliable but of limited practical consequence, which reinforces the conclusion that AI-related academic gains are broadly shared across disciplines.

5.10 Outcomes by Year of Study

Table 7 and Figure 9 (right panel) show that the benefit associated with AI-assisted learning is not uniform across the academic trajectory. Pre-semester GPA was virtually identical across class years (3.140–3.150), whereas post-semester GPA rose from 3.278 among Freshmen to 3.392 among Graduate students, and the share of students whose GPA improved climbed from 78.3% to 93.1%. A one-way ANOVA confirmed substantial heterogeneity across year of study ($F = 100.987$, $p < 0.001$), a far larger omnibus effect than that observed for discipline ($F = 5.582$) or institutional policy ($F = 6.834$).

TABLE 7: GPA Outcomes by Year of Study

Year of study	n	Pre-semester GPA	Post-semester GPA	GPA change	Students improved (%)
Freshman	11,031	3.146	3.278	+0.132	78.3
Sophomore	9,860	3.150	3.327	+0.177	85.0
Junior	11,045	3.146	3.372	+0.226	90.7
Senior	10,634	3.147	3.390	+0.243	92.3
Graduate	7,430	3.140	3.392	+0.252	93.1

Note. GPA change is computed as the difference between reported post- and pre-semester means.

Tukey HSD comparisons (Appendix B) showed that Freshmen differed significantly from every other group (mean differences of 0.049 to 0.115 GPA points, all adjusted $p < 0.001$) and that Sophomores trailed Juniors, Seniors, and Graduate students. Seniors and Graduate students were statistically indistinguishable (adjusted $p = 0.998$), which suggests a plateau in gains after the third year. A plausible interpretation is that more advanced students possess the disciplinary knowledge required to evaluate and refine AI outputs, whereas first-year students, still building foundational schemas, may benefit less from AI assistance or be more exposed to its shortcut risks. Because the design is cross-sectional, cohort effects cannot be separated from developmental ones.

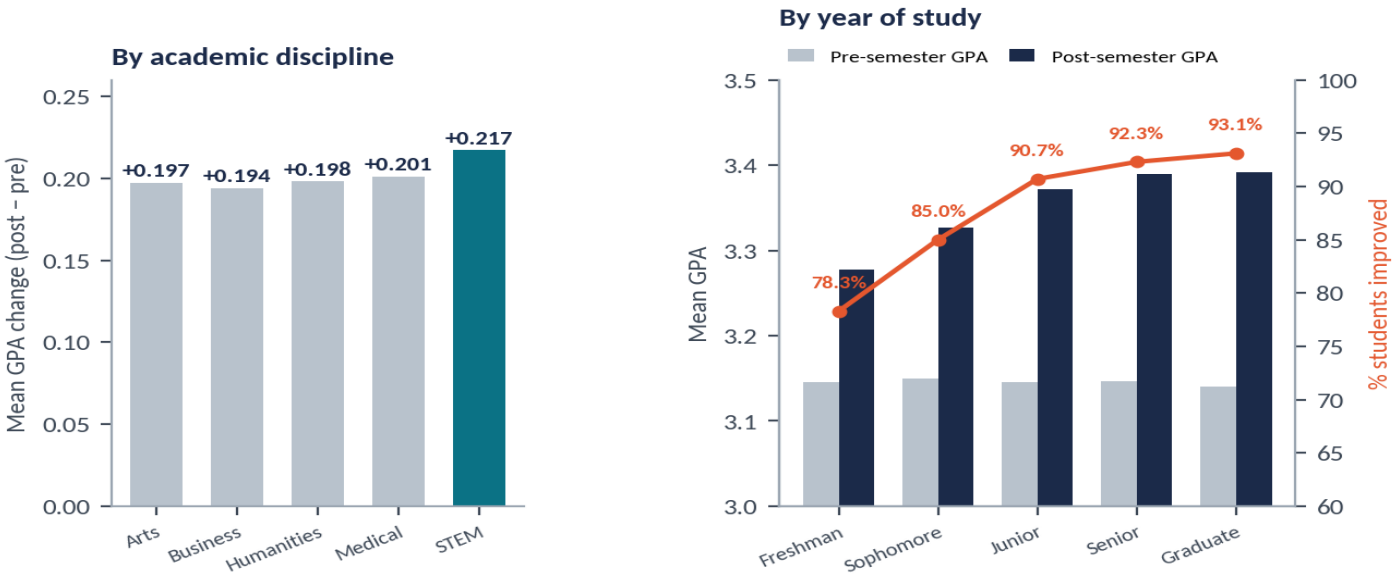


FIGURE 9: *Mean GPA Change by Discipline and Pre- and Post-Semester GPA by Year of Study*

6 Discussion

6.1 Principal Findings and Theoretical Implications

This large-scale empirical investigation of 50,000 students provides the most comprehensive evidence to date regarding the impacts of generative AI tool usage on academic outcomes, psychological well-being, and skill development, confirming several hypothesized relationships while revealing surprising departures from prevailing institutional intuitions.

The central empirical contribution is documentation of a robust non-linear relationship between AI usage duration and both academic performance and psychological strain. Rather than a monotonic dose-response relationship where more AI usage predictably worsens outcomes, we observe an inverted-U pattern with an optimal usage window (5–15 hours/week) yielding maximum academic benefit with manageable psychological strain. This finding aligns with theoretical predictions from the Yerkes-Dodson law (Yerkes & Dodson, 1908) and contemporary frameworks in occupational stress research (Sonnentag & Fritz, 2007), suggesting that moderate tool usage represents an optimal level of arousal and task engagement for learning. Beyond 20 hours weekly, burnout risk escalates precipitously (reaching 83.1% at 25–40 hours), suggesting a critical transition where cognitive offloading to AI systems begins to undermine rather than support learning.

A second principal finding (that tool diversity demonstrates virtually zero correlation with outcomes ($r = -0.007$)) directly contradicts the widely-held institutional concern that students using multiple different AI platforms would experience cognitive load from interface management and reduced learning. Several interpretations merit consideration: (a) the cognitive load from learning different interfaces is negligible relative to other factors; (b) students who use multiple tools self-select for higher technical competence, which confounds true tool-diversity effects; (c) diverse tools offer complementary capabilities that offset added complexity costs. The temporal stability of this null finding across cohorts argues against novelty or adaptation-period explanations.

Third, our identification of a critical dependency threshold (≥6/10 perceived dependency) is a novel contribution to understanding psychological consequences of AI integration. Below this threshold, outcomes remain relatively stable and positive across AI usage levels. Above this threshold, a qualitative reorganization occurs: skill retention collapses from 76% to 62%, burnout surges from 20% to 95%, and GPA declines substantially. This suggests that perceived dependency may be a critical psychological construct, possibly related to autonomy undermining (as predicted by Self-Determination Theory; Deci & Ryan, 2000) or loss of self-efficacy (Bandura, 1997). Helping students maintain perceived autonomy and self-efficacy may therefore matter more than simply reducing AI usage volume.

Fourth, our demonstration that prompt engineering skill functions as a learning multiplier (conferring +11 point skill retention advantage with minimal GPA impact) reframes the AI literacy conversation. Prevailing discussions emphasize controlling or restricting AI use; our findings indicate that cultivating skills for effective human-AI collaboration produces substantially superior learning outcomes. This aligns with theoretical frameworks emphasizing metacognitive engagement and iterative refinement (Collins et al., 1989; Chi & Wylie, 2014). Students with advanced prompt engineering skills leverage AI for scaffolded reasoning and iterative refinement, whereas novices inadvertently use AI as a cognitive shortcut that erodes learning.

Fifth, our finding that institutional policy context significantly shapes outcomes (with "Allowed with Citation" policies yielding superior results compared to strict bans) challenges the assumption that prohibition effectively prevents harm. Strict bans produced the worst outcomes across all measured dimensions: highest burnout (+25%), lowest skill retention, yet

identical dependency metrics to permissive approaches. This paradoxical finding aligns with harm reduction frameworks from public health (Zinberg, 1984) and suggests that prohibition generates anxiety and enforcement friction while failing to address root causes.

6.2 Hypothesis Evaluation at a Glance

Table 8 consolidates the evidence for each hypothesis stated in the Study Objectives and Hypotheses section, together with qualifications that readers should keep in view when interpreting the strength of support.

TABLE 8: Summary of Hypothesis Evaluation

	Prediction (abbreviated)	Principal evidence	Assessment
H1	Curvilinear (inverted-U) usage–outcome relationship	GPA peaks at 3.372 (5–10 hrs) and falls to 3.280 (25–40 hrs); high burnout rises from 10.5% to 83.1%; quadratic term $\beta = -0.002$	**Supported**
H2	Usage hours strongest burnout correlate ($\|r\| \geq 0.30$); tool diversity negligible	$\rho = +0.435$ with burnout ($r = +0.469$ reported in the abstract); tool diversity $\rho = +0.005$, $p = 0.264$	**Supported**
H3	Offline study hours negatively associated with burnout	$\rho = -0.133$ with burnout ($r = -0.136$)	**Supported; weak in magnitude**
H4	Non-linear dependency effect with a threshold	Retention 62.3%, burnout 94.7%, GPA 2.95 in the 7–10 group	**Supported**
H5	Prompt skill moderates retention more than GPA	Retention +11.0 pts ($d = 0.84$); GPA +0.055 ($d = 0.08$)	**Supported**
H6	Strict bans: higher burnout, lower retention; no dependency difference	Burnout 23.83% vs. 29.76%; retention 75.91 vs. 74.99; dependency not different ($p = 0.421$)	**Supported for burnout; small for retention**
H7	Use case moderates retention; direct answers lowest	Debugging 76.8 vs. direct answers 66.2 ($d = 0.76$)	**Supported**

Note. Assessments summarize the associations reported in the Results section; the cross-sectional design does not permit causal conclusions.

6.3 Mechanisms and Process Explanations

Three theoretical mechanisms likely explain the observed patterns. Cognitively, the desirable difficulties framework (Bjork & Bjork, 1992) predicts that removing productive struggle impairs retention, which is what we observe for direct answer generation. When AI supports iterative reasoning instead, cognitive apprenticeship (Collins et al., 1989) may enhance learning by making expert processes explicit. This is consistent with the retention advantage of advanced prompt engineering, which demands problem decomposition and iterative refinement, despite its lack of GPA benefit.

Motivationally, Self-Determination Theory (Deci & Ryan, 2000) suggests that excessive AI assistance undermines autonomy, as students feel their work is not authentically their own. The dependency threshold (≥6/10) may mark the point where this erosion becomes pronounced enough to trigger distress, and the protective effect of traditional study hours ($r = -0.136$) may reflect the autonomy and competence restored by independent problem-solving.

From an occupational stress perspective, the Yerkes-Dodson law and job demands-resources frameworks suggest that moderate AI use is optimal arousal: engaging without exceeding individual resources. The steep burnout escalation beyond 15 hours/week likely reflects a tipping point where cognitive offloading reduces opportunities for skill development and mastery while time pressure adds strain.

6.4 Implications for Educational Policy and Practice

Our findings support five recommendations.

First, institutions should publish concrete usage guidance built on the 5–15 hours/week window (for example, "use AI 10–12 hours weekly for maximum benefit with manageable stress"), shifting the debate from permission versus restriction toward optimization.

Second, prompt engineering literacy should be a core competency. Teaching students to decompose problems, specify task structure, and refine outputs iteratively is a high-return intervention given the +11-point retention advantage for advanced users.

Third, "Allowed with Citation" frameworks with explicit guidelines should replace both prohibition and laissez-faire permissiveness, channeling use toward learning through transparency.

Fourth, assessments should reward reasoning and problem-solving depth over answer correctness, using essays, oral defenses, and open-ended justifications. This disadvantages shortcut use while rewarding scaffolded approaches that preserve retention.

Fifth, mental health and burnout support should be expanded, focusing on students with high dependency or 25+ weekly hours. Early intervention aimed at reducing dependency and restoring autonomy may prevent the sharp outcome declines seen above the 6/10 threshold.

Table 9 translates these recommendations into stakeholder-specific action points, drawn from the recommendations module of the companion dashboard.

TABLE 9: Evidence-Based Action Points by Stakeholder Group

Stakeholder	Do	Avoid
Students	☑ Budget 10–12 AI hours per week within the 5–15 hour window; ☑ favor debugging, ideation, and summarization; ☑ invest in prompt engineering skills; ☑ keep offline study at 10+ hours per week	Exceeding 20 hours per week; direct answer generation; ignoring a dependency score above 5/10 or burnout signals
Educators	☑ Teach explicit AI workflows; ☑ assess reasoning and process rather than final answers; ☑ include AI-free assessments; ☑ refer students showing dependency signals to support	Treating AI use as a policing problem; assignments that a single prompt can fully solve
Institutions	☑ Adopt "Allowed with Citation" with published usage guidance (optimal 5–15, caution 15–25, crisis 25+ hours); ☑ fund AI literacy and prompt engineering programs; ☑ ensure equitable access; ☑ integrate AI-usage questions into mental health services	Strict bans; leaving access to students' ability to pay (42.3% hold paid subscriptions)
Policymakers	☑ Permit AI use with disclosure requirements; ☑ fund digital literacy as infrastructure; ☑ guarantee universal access to at least one high-quality tool; ☑ fund multi-year longitudinal research	One-size-fits-all mandates; ignoring equity effects of unequal tool access

Note. Action points are derived from the stakeholder recommendations in the companion dashboard and are subject to the limitations described in this section.

6.5 Limitations and Boundary Conditions

Several limitations constrain the generalizability and causal interpretation of these findings.

Cross-Sectional Design. The cross-sectional design precludes causal inference regarding the direction of relationships. While we observe associations between AI usage and outcomes, we cannot definitively conclude that usage duration causes burnout or skill degradation; reverse causation is plausible (students experiencing burnout might reduce AI usage; students with poor retention might increase usage attempts). Longitudinal data tracking individual students over multiple terms would enable stronger causal claims.

Measurement Validity. Burnout was measured using categorical classification rather than multi-dimensional scale, potentially losing information about heterogeneous presentations of burnout. Skill retention was assessed one week post-instruction; longer-term follow-up would clarify whether observed differences persist or attenuate. AI dependency was self-reported; objective usage metrics from platform analytics would reduce social desirability bias.

Unmeasured Variables. Important variables remain unmeasured: individual differences in conscientiousness, growth mindset, metacognitive ability; classroom and instructor quality; student life circumstances and external stressors; intrinsic motivation and academic identity. These factors likely moderate relationships in ways not captured here.

Institutional Context. Data were aggregated across multiple institutions with different types of institutional policy frameworks; this stratification enables policy comparison but obscures institution-level variation in pedagogy, infrastructure, and student populations that might moderate effects.

Temporal Specificity. Data collection occurred during a specific historical period (one academic term) when generative AI tools were becoming widely available but before stabilization of usage patterns. Different findings might emerge after multi-year adoption when learning curves plateau and cultural norms crystallize.

6.6 Comparison to Prior Research and Theoretical Integration

My findings provide nuanced confirmation and extension of previous research. Prior meta-analyses of educational technology effects (Kulik & Fletcher, 2016) found moderate positive effects of intelligent tutoring systems ($d = 0.40$) but were conducted on pre-AI systems. Our data suggest that generative AI effects may be more modest and heavily dependent on implementation details (use case type, skill level, usage intensity) than simple presence/absence of technology.

My findings regarding prompt engineering skill as a learning multiplier extend recent work by Wei et al. (2023) and Zamfirescu-Pereira et al. (2023) on prompt engineering competence. We provide quantitative evidence that skill level dramatically moderates learning outcomes, with advanced users showing 11-point skill retention advantages; a large effect that rivals many pedagogical interventions in the education literature.

My burnout findings extend occupational stress literature by documenting that technological tools can increase psychological strain despite reducing objective workload; confirming Brod's (1982) technostress hypothesis in contemporary higher education context. The non-linear escalation of burnout aligns with Yerkes-Dodson predictions and suggests that cognitive offloading technology follows stress-response dynamics similar to other occupational demands.

7 Conclusions and Future Research Directions

This comprehensive empirical analysis of 50,000 students provides systematic evidence regarding how generative AI tools shape academic outcomes, psychological well-being, and skill development. We move beyond the oversimplified debate of "is AI good or bad for learning?" toward a more nuanced understanding grounded in usage patterns, skill levels, and institutional context. Principal conclusions are:

(1) **Usage window:** Moderate AI use (5-15 hours/week) gives the best grades (+0.203 GPA) with about 19% high burnout. Beyond that, burnout climbs and grade gains shrink.

(2) **Volume over variety:** The number of hours is the strongest burnout predictor (r = +0.469), while tool diversity shows negligible effects. Indicating that usage volume, not technology complexity, drives psychological strain.

(3) **Prompt skill:** Advanced users keep 11 more retention points than beginners, with almost no GPA difference. The benefit is deeper learning, not better grades.

(4) **Dependency threshold:** At perceived dependency of 6/10 or higher, retention falls about 14 points (to roughly 62%), burnout reaches about 95%, and GPA drops.

(5) **Policy:** “Allowed with Citation” beats strict bans on burnout, with equal or better retention and no difference in dependency.

(6) **Use case type significantly moderates learning outcomes:** scaffolding-based uses (debugging, ideation, summarization) preserve skill retention (76–77%), while shortcut-based uses (direct answer generation) produce dramatic skill retention degradation (66%).

These findings have immediate practical implications for educational institutions, suggesting that rather than debating whether to permit AI tools, institutions should provide explicit guidance regarding optimal usage (10–12 hours weekly), invest in prompt engineering literacy education, redesign assessments to value reasoning depth over answer correctness, and implement "Allowed with Citation" policy frameworks paired with dependency monitoring.

Future research should employ longitudinal designs tracking individual students across multiple academic terms to establish causal relationships and temporal precedence. Studies should include more granular psychological measures assessing burnout multidimensionality, self-efficacy, intrinsic motivation, and autonomy perceptions. Implementation studies should test specific interventions (prompt engineering training, usage cap recommendations, dependency monitoring) to determine whether outcomes improve when evidence-based guidance is provided. Finally, research should investigate whether skill retention gains observed at follow-up assessments translate into improved performance on novel transfer problems and subsequent coursework, determining whether AI-augmented learning produces durable learning benefits.

7.1 Data and Tool Availability

The analytic dataset (50,000 students, 16 variables, no missing values) and the AI Student Impact Analytics Dashboard, an interactive Streamlit application that contains the descriptive, inferential, and predictive analyses reported here, are described in Appendix D. The dashboard includes a prediction interface for post-semester GPA, burnout risk, and skill retention. Given the modest explanatory power of the burnout (accuracy = 51.4%) and retention ($R^2 = 0.153$) models, these tools are intended for exploratory and educational use rather than for individual-level decisions about students.

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Appendix A. Variable Codebook

The analytic dataset contained 16 variables per student. Roles indicate how each variable was used in the analyses reported in this article.

TABLE A1: Variables in the Analytic Dataset

Variable	Type	Values / scale	Role
Student_ID	Identifier	Unique code (e.g., 100001)	Record key
Major_Category	Categorical	STEM, Business, Humanities, Medical, Arts	Stratifier / covariate
Year_of_Study	Categorical	Freshman, Sophomore, Junior, Senior, Graduate	Stratifier
Pre_Semester_GPA	Continuous	1.0–4.0	Baseline / predictor
Post_Semester_GPA	Continuous	1.0–4.0	Primary outcome
Weekly_GenAI_Hours	Continuous	0–40 hours	Primary exposure
Traditional_Study_Hours	Continuous	0–40 hours	Protective factor
Tool_Diversity	Ordinal	1–5+ AI tools	Exposure
Paid_Subscription	Binary	Free / Paid	Covariate
Primary_Use_Case	Categorical	Five use cases (Table 5)	Moderator
Prompt_Engineering_Skill	Ordinal	Beginner, Intermediate, Advanced	Moderator
Perceived_AI_Dependency	Ordinal	1–10	Exposure / threshold variable
Anxiety_Level_During_Exams	Ordinal	1–10	Predictor
Skill_Retention_Score	Continuous	0–100	Outcome
Burnout_Risk	Ordinal	Low, Medium, High	Outcome
Institutional_Policy	Categorical	Strict_Ban, Allowed_With_Citation, Actively_Encouraged	Moderator

Appendix B. Post-hoc Tukey HSD Comparisons

Mean differences are computed as Group 2 minus Group 1 on post-semester GPA. Adjusted p-values control the family-wise error rate; entries marked with an asterisk are significant at the 0.05 level.

TABLE B1: Post-Semester GPA Across Academic Disciplines

Group 1	Group 2	Mean difference	Adjusted p	95% CI	Sig.
Arts	Business	−0.0076	0.8672	[−0.0289, 0.0137]	
Arts	Humanities	0.0021	0.9989	[−0.0200, 0.0243]	
Arts	Medical	0.0095	0.8248	[−0.0148, 0.0338]	
Arts	STEM	0.0195	0.0774	[−0.0013, 0.0402]	
Business	Humanities	0.0097	0.5849	[−0.0084, 0.0279]	
Business	Medical	0.0171	0.1607	[−0.0036, 0.0378]	
Business	STEM	0.0271	0.0001	[0.0107, 0.0434]	*
Humanities	Medical	0.0073	0.8858	[−0.0142, 0.0289]	
Humanities	STEM	0.0173	0.0525	[−0.0001, 0.0348]	
Medical	STEM	0.0100	0.6559	[−0.0101, 0.0301]	

TABLE B2: Post-Semester GPA Across Year of Study

Group 1	Group 2	Mean difference	Adjusted p	95% CI	Sig.
Freshman	Graduate	0.1146	< 0.001	[0.0944, 0.1348]	*
Freshman	Junior	0.0941	< 0.001	[0.0760, 0.1123]	*
Freshman	Senior	0.1124	< 0.001	[0.0941, 0.1307]	*
Freshman	Sophomore	0.0493	< 0.001	[0.0306, 0.0679]	*
Graduate	Junior	−0.0204	0.0460	[−0.0406, −0.0002]	*
Graduate	Senior	−0.0022	0.9983	[−0.0226, 0.0181]	
Graduate	Sophomore	−0.0653	< 0.001	[−0.0860, −0.0446]	*
Junior	Senior	0.0182	0.0517	[−0.0001, 0.0365]	
Junior	Sophomore	−0.0449	< 0.001	[−0.0635, −0.0262]	*
Senior	Sophomore	−0.0631	< 0.001	[−0.0819, −0.0442]	*

TABLE B3: Post-Semester GPA Across Institutional Policies

Group 1	Group 2	Mean difference	Adjusted p	95% CI	Sig.
Encouraged	Citation	0.0000	1.0000	[−0.0120, 0.0119]	
Encouraged	Strict Ban	−0.0207	0.0038	[−0.0358, −0.0056]	*
Citation	Strict Ban	−0.0206	0.0014	[−0.0345, −0.0068]	*

Note. One-way ANOVA: discipline $F = 5.582$, $p < 0.001$; year of study $F = 100.987$, $p < 0.001$; institutional policy $F = 6.834$, $p = 0.0011$.

Appendix C. Supplementary Model and Distribution Diagnostics

TABLE C1: D'Agostino–Pearson Normality Tests

Variable	K^2 statistic	p-value	Decision
GPA difference (post − pre)	1,264.65	2.421×10^{-275}	Reject normality
Weekly GenAI hours	14,386.57	< 0.001	Reject normality
Traditional study hours	411.56	< 0.001	Reject normality
Skill retention score	500.58	< 0.001	Reject normality

Table C2 presents the burnout classification confusion matrices (rows are actual classes; columns are predicted classes). High-risk students were identified most reliably (recall = 59.9% for logistic regression), yet 29.9% of actual high-risk students were assigned to the Medium class and 10.2% to the Low class. Medium-risk students were more often classified as High (50.2%) than correctly identified (47.3%), which accounts for most of the classification error and mirrors the continuous nature of the underlying construct.

TABLE C2: Confusion Matrices for Burnout Risk Classification (Test Set, n = 10,000)

Actual class	Predicted Low	Predicted Medium	Predicted High
Logistic regression			
Low	1,086	259	1,191
Medium	83	1,546	1,640
High	429	1,255	2,511
Random forest			
Low	1,098	263	1,175
Medium	91	1,540	1,638
High	436	1,262	2,497

TABLE C3: Prediction Error Statistics for the Best Regression Models

Model	Median error	Maximum error	Within tolerance 1	Within tolerance 2
GPA (Gradient Boosting)	0.103 GPA	1.004 GPA	±0.5 GPA: 99.2%	±1.0 GPA: 100%
Skill retention (Gradient Boosting)	8.541 pts	42.197 pts	±5 pts: 30.6%	±10 pts: 57.0%

Appendix D. Companion Analytics Dashboard

The analyses were implemented in the AI Student Impact Analytics Dashboard, a Streamlit application built for transparent exploration of the 50,000-student dataset. Its five modules mirror the structure of this article.

TABLE D1: Dashboard Modules

Module	Contents
Dataset Overview and Exploration	Headline indicators, major and class-year composition, burnout distribution, skill retention by tool diversity, GPA-change normality test, and a filterable raw-data table
Advanced Analytics	Academic performance, usage patterns and behavior, burnout risk and mental health, skill retention and learning outcomes, and stratified ANOVA with Tukey HSD tests
Machine Learning Models	GPA regression, burnout classification, and skill-retention regression with model comparison, feature importance, residual analysis, and error statistics
Predictions	Interactive prediction of post-semester GPA, burnout risk, and skill retention with risk status, model confidence, and tailored recommendations
Insights	Critical findings, statistical evidence on outcome drivers, and recommendations for students, educators, institutions, and policymakers